STAT 37400
Nonparametric Inference

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Foundations and Histogram Estimation 1

1.1 Setup and Decision Theory

Notation 1.1.1.

There is an unknown data-generating distribution P. We observe i.i.d. data

$$Z_1, \ldots, Z_n \overset{\text{iid}}{\sim} P.$$

We have two common regimes:

1. *Unsupervised:* $Z_i = X_i$ with $X_i \in \mathbb{R}^p$.
2. *Supervised (regression):* $Z_i = (X_i, Y_i)$ with $X_i \in \mathbb{R}^p$ and $Y_i \in \mathbb{R}$.

A generic nonparametric workflow is the following:

1. Decide what property of P you care about (a *functional* $\theta(P)$).
2. Decide how to measure performance (statistical decision theory: loss/risk).
3. Ask for the best possible performance under assumptions (information-theoretic lower bounds).
4. Propose an estimator and check whether it achieves the benchmark.

Definition 1.1.2 (Statistical functional / parameter).

A *parameter* is a map $\theta : \mathscr{P} \to \Theta$ that assigns to each distribution P a quantity $\theta(P)$ (a number, vector, or function) we want to learn from data.

Example 1.1.3 (Common functionals (one-dimensional Y)).

Suppose $Z_i = Y_i \in \mathbb{R}$ i.i.d. $\sim P$.

1. Mean: $\theta(P) = \mathbb{E}_P[Y_i]$ (assuming $\mathbb{E}_P[|Y_i|] < \infty$).
2. Variance: $\theta(P) = \mathbb{E}_P[(Y_i - \mathbb{E}_P[Y_i])^2]$ (assuming $\mathbb{E}_P[Y_i^2] < \infty$).

3. Median:
$$\theta(P) = \inf\{y \in \mathbb{R} : \ \mathbb{P}_P(Y_i \le y) \ge 1/2\}.$$
4. CDF at a point y (or as a whole function $y \mapsto F(y)$):
$$\theta(P) = F(y) \coloneqq \mathbb{P}_P(Y_i \le y) = \mathbb{E}_P[\mathbf{1}\{Y_i \le y\}].$$
5. Tail conditional (extreme values): for fixed y, t,
$$\theta(P) = \mathbb{P}_P(Y_i \ge y \mid Y_i \ge t).$$
6. Density: if P has density f, then one may want $\theta(P) = f$ (a function), or a pointwise target $\theta(P) = f(y_0)$ for fixed $y_0 \in \mathbb{R}$.

Definition 1.1.4 (Estimator).

An estimator of $\theta(P)$ is any measurable function of the data,
$$\hat{\theta} = \hat{\theta}(Z_1, \dots, Z_n).$$

Definition 1.1.5 (Squared-error risk / MSE).

For real-valued $\theta(P)$ and $\hat{\theta}$, define the (mean squared) risk under P as
$$R_P(\hat{\theta}, \theta) \coloneqq \mathbb{E}_P\left[(\hat{\theta} - \theta(P))^2\right],$$
assuming $\mathbb{E}_P[\hat{\theta}^2] < \infty$.

Lemma 1.1.6 (Bias–variance decomposition).

Let $\theta \in \mathbb{R}$ and $\hat{\theta}$ be any real-valued random variable with $\mathbb{E}[\hat{\theta}^2] < \infty$. Then
$$\mathbb{E}\left[(\hat{\theta} - \theta)^2\right] = \mathrm{Var}(\hat{\theta}) + \left(\mathbb{E}[\hat{\theta}] - \theta\right)^2.$$

Proof of Lemma 1.1.6. Write $\hat{\theta} - \theta = (\hat{\theta} - \mathbb{E}\hat{\theta}) + (\mathbb{E}\hat{\theta} - \theta)$ and expand the square; the cross term is $2\,\mathbb{E}\left[\hat{\theta} - \mathbb{E}[\hat{\theta}]\right] \cdot (\mathbb{E}[\hat{\theta}] - \theta) = 0$.

Remark 1.1.7. In many nonparametric problems, you *cannot* make both variance and bias small at the same time. The tuning parameter is chosen to balance $\mathrm{Var}(\hat{\theta})$ against $\mathrm{Bias}(\hat{\theta})^2$.

Example 1.1.8 (Sample mean).

Let $\theta(P) = \mathbb{E}_P[Y_i]$ and consider $\hat{\theta} = \bar{Y} := \frac{1}{n}\sum_{i=1}^n Y_i$. Then $\mathbb{E}_P[\bar{Y}] = \theta(P)$ (unbiased), and if $\mathrm{Var}_P(Y_i) = \sigma^2 < \infty$,

$$R_P(\bar{Y}, \theta) = \mathrm{Var}_P(\bar{Y}) = \frac{\sigma^2}{n}.$$

This is the canonical $1/n$ MSE rate.

Definition 1.1.9 (Worst-case and minimax risk).

Fix a class $\mathscr{P}$ of distributions. The *worst-case risk* of $\hat{\theta}$ over $\mathscr{P}$ is

$$\sup_{P \in \mathscr{P}} R_P(\hat{\theta}, \theta) = \sup_{P \in \mathscr{P}} \mathbb{E}_P\big[(\hat{\theta} - \theta(P))^2\big].$$

The *minimax risk* over $\mathscr{P}$ is

$$\inf_{\hat{\theta}} \sup_{P \in \mathscr{P}} \mathbb{E}_P\big[(\hat{\theta} - \theta(P))^2\big],$$

where the infimum ranges over all estimators based on $(Z_1, \ldots, Z_n)$.

We can think of minimax risk in the context of a game: the statistician chooses $\hat{\theta}$; nature chooses $P \in \mathscr{P}$. The minimax value is the best achievable worst-case performance.

Example 1.1.10.

If $\mathscr{P} = \{P : \mathrm{Var}_P(Y_i) < \infty\}$, then for $\hat{\theta} = \bar{Y}$,

$$\sup_{P \in \mathscr{P}} R_P(\bar{Y}, \theta) = \sup_{P \in \mathscr{P}} \frac{\mathrm{Var}_P(Y_i)}{n} = \infty,$$

so this "model class" is too broad to yield a meaningful worst-case guarantee. We can restrict this class of distributions to ones with bounded variance. Fix $M < \infty$ and now let $\mathscr{P}_M = \{P : \mathrm{Var}_P(Y_i) \le M\}$. Then

$$\sup_{P \in \mathscr{P}_M} R_P(\bar{Y}, \theta) = \frac{M}{n}.$$

Moreover (nontrivial fact): $\bar{Y}$ is minimax optimal on $\mathscr{P}_M$, i.e.

$$\inf_{\hat{\theta}} \sup_{P \in \mathscr{P}_M} \mathbb{E}_P\big[(\hat{\theta} - \mathbb{E}_P Y_i)^2\big] = \frac{M}{n}.$$

1.2 Empirical Methods

Example 1.2.1.

Fix $y \in \mathbb{R}$ and the target $\theta(P) = F(y) = \mathbb{P}_P(Y_i \le y)$. Define

$$\hat{F}_n(y) \coloneqq \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{Y_i \le y}.$$

Then $\mathbb{E}_P[\hat{F}_n(y)] = F(y)$ (unbiased), and since $\mathbf{1}\{Y_i \le y\} \sim \text{Bernoulli}(F(y))$,

$$\text{Var}_P(\hat{F}_n(y)) = \frac{F(y)\big(1 - F(y)\big)}{n} \le \frac{1}{4n}.$$

In particular, even if $\mathscr{P}$ is the class of *all* distributions, the worst-case MSE is at most $1/(4n)$.

In many parametric problems, allowing a small bias does not change the canonical rate: if $n \cdot \text{Bias}(\hat{\theta})^2 \to 0$, then bias is negligible relative to variance at the usual $1/n$ scale. In harder (typically nonparametric) problems, bias often dominates unless handled carefully.

A common strategy is to start with a family $\{\hat{\theta}_\lambda : \lambda \in \Lambda\}$ and choose the indexing tuning parameter λ to balance

$$\mathbb{E}_P[(\hat{\theta}_\lambda - \theta(P))^2] = \text{Var}_P(\hat{\theta}_\lambda) + \text{Bias}_P(\hat{\theta}_\lambda)^2.$$

Example 1.2.2 (Histogram estimator at a point).

Suppose $Y_i \in [0,1]$ and P has density f. Fix $y_0 \in [0,1]$ and the target $\theta(P) = f(y_0)$. Partition $[0,1]$ into K bins $I_1, \ldots, I_K$ of equal length h (so $h = 1/K$), and let I_j be the bin containing y_0. Define the histogram estimator

$$\hat{f}_h(y_0) \coloneqq \frac{\#\{i :\ Y_i \in I_j\}}{nh} = \frac{1}{nh}\sum_{i=1}^{n} \mathbf{1}\{Y_i \in I_j\}.$$

Here the tuning parameter is $\lambda = h$.

Remark 1.2.3 (Variance increases as h decreases). Let $p_j \coloneqq \mathbb{P}_P(Y_i \in I_j)$. Since $\mathbf{1}\{Y_i \in I_j\} \sim \text{Bernoulli}(p_j)$,

$$\text{Var}_P\big(\hat{f}_h(y_0)\big) = \frac{1}{nh^2}\, p_j(1 - p_j) \le \frac{1}{nh^2}\, p_j.$$

If we assume a uniform bound $f(y) \le M < \infty$ for all y, then

$$p_j = \int_{I_j} f(y)\, dy \le M|I_j| = Mh,$$

so

$$\text{Var}_P\big(\hat{f}_h(y_0)\big) \le \frac{M}{nh}.$$

Remark 1.2.4 (A typical bias bound and the implied rate (heuristic)). Suppose (under additional smoothness assumptions on f) that the histogram bias satisfies

$$\left|\mathrm{Bias}_P(\hat{f}_h(y_0))\right| = \left|\mathbb{E}_P\left[\hat{f}_h(y_0)\right] - f(y_0)\right| \le Ch$$

for some constant C. Then

$$\mathbb{E}_P\left[(\hat{f}_h(y_0) - f(y_0))^2\right] \le \frac{M}{nh} + C^2h^2.$$

Minimizing the RHS over h suggests $h^* \asymp n^{-1/3}$ and an MSE rate $\asymp n^{-2/3}$, which is slower than the $1/n$ rate seen in simpler problems (e.g. mean/CDF-at-a-point).

Aside 1.2.5. The specific bias bound $O(h)$ is *not automatic*; it encodes smoothness of f near y_0. The point of the example is the mechanism: shrinking h reduces bias but inflates variance, and the optimal choice balances the two.

1.3 Histogram Analysis

Assume $Y_i \in [0,1]$ and P admits a density f (with respect to Lebesgue measure). Fix $y_0 \in [0,1]$ and target the pointwise value $\theta(P) = f(y_0)$.

Definition 1.3.1 (Histogram estimator at y_0).

Partition $[0,1]$ into K intervals (bins) $I_1, \dots, I_K$ of equal length $h \coloneqq 1/K$. Let I_j be the unique bin containing y_0. Define

$$\hat{f}_h(y_0) \coloneqq \frac{\#\{i : Y_i \in I_j\}}{nh} = \frac{1}{nh}\sum_{i=1}^{n} \mathbf{1}\{Y_i \in I_j\}.$$

The tuning parameter is the bin width h.

Assumption 1.3.2 (Uniform boundedness).

We assume there exists $M < \infty$ such that $f(y) \le M$ for all $y \in [0,1]$.

Lemma 1.3.3 (Variance bound).

Under assumption 1.3.2,

$$\mathrm{Var}_P\left(\hat{f}_h(y_0)\right) \le \frac{M}{nh}.$$

Proof of Lemma 1.3.3. Let $p_j \coloneqq \mathbb{P}_P(Y_i \in I_j) = \int_{I_j} f(y)\,dy \le Mh$. Since $\mathbf{1}\{Y_i \in I_j\} \sim \mathrm{Bernoulli}(p_j)$,

$$\mathrm{Var}_P(\hat{f}_h(y_0)) = \mathrm{Var}_P\Big(\frac{1}{nh}\sum_{i=1}^{n}\mathbf{1}\{Y_i \in I_j\}\Big) = \frac{1}{nh^2}\,p_j(1-p_j) \le \frac{1}{nh^2}\,p_j \le \frac{M}{nh}.$$

Recall that the distribution P does not change if we modify f at a single point. In particular, $f(y_0)$ is not identified from P unless we restrict the class of densities (e.g. by continuity). This motivates smoothness assumptions.

Definition 1.3.4 (L-Lipschitz continuity).

A function $f : [0,1] \to \mathbb{R}$ is *L-Lipschitz* if

$$|f(y) - f(y')| \le L|y - y'| \qquad \text{for all } y, y' \in [0,1].$$

Assumption 1.3.5 (Lipschitz smoothness).

We assume the density f is L-Lipschitz on $[0,1]$.

Lemma 1.3.6 (Bias expression).

Let $I_j = (a_j, b_j]$ be the bin containing y_0, so $b_j - a_j = h$. Then

$$\mathbb{E}_P[\hat{f}_h(y_0)] = \frac{1}{h}\mathbb{P}_P(Y_i \in I_j) = \frac{F(b_j) - F(a_j)}{h}.$$

In particular, under assumption 1.3.5,

$$\left|\mathbb{E}_P[\hat{f}_h(y_0)] - f(y_0)\right| \le Lh.$$

Proof of Lemma 1.3.6. We see $\mathbb{E}_P[\hat{f}_h(y_0)] = (F(b_j) - F(a_j))/(b_j - a_j)$ is the slope of F over (a_j, b_j). By the mean value theorem (since $F' = f$ wherever differentiable), there exists $\xi \in (a_j, b_j)$ such that

$$\frac{F(b_j) - F(a_j)}{b_j - a_j} = f(\xi).$$

Therefore

$$\left|\mathbb{E}_P[\hat{f}_h(y_0)] - f(y_0)\right| = |f(\xi) - f(y_0)| \le L|\xi - y_0| \le Lh.$$

Now, let $\mathscr{P}(L, M)$ be the class of distributions on $[0,1]$ with densities f satisfying assumptions 1.3.2 and 1.3.5.

Proposition 1.3.7 (Worst-case MSE upper bound).

For every $h \in (0,1)$,

$$\sup_{P \in \mathscr{P}(L,M)} \mathrm{MSE}_P\big(\hat{f}_h(y_0)\big) \leq \frac{M}{nh} + L^2 h^2.$$

Proof of Proposition 1.3.7. Combine Lemma 1.3.3 with Lemma 1.3.6 and the bias–variance decomposition.

Remark 1.3.8 (Consistency conditions). From Proposition 1.3.7, it is enough that $h \to 0$ and $nh \to \infty$ to force both terms to 0.

Ignoring constants, the upper bound in Proposition 1.3.7 is minimized by choosing

$$h \asymp n^{-1/3},$$

which yields the *minimax optimal rate under the Lipschitz assumption*

$$\sup_{P \in \mathscr{P}(L,M)} \mathrm{MSE}_P\big(\hat{f}_h(y_0)\big) \lesssim n^{-2/3}.$$

Remark 1.3.9. This is a canonical nonparametric phenomenon: the pointwise density problem is harder than estimating $F(y_0)$, and $n^{-2/3}$ is slower than $1/n$.

Theorem 1.3.10 (Minimax rate for pointwise density estimation).

There exist constants $c(L,M), C(L,M) > 0$ such that for all n,

$$c(L,M)\, n^{-2/3} \leq \inf_{\tilde{f}(y_0)} \sup_{P \in \mathscr{P}(L,M)} \mathbb{E}_P\big[(\tilde{f}(y_0) - f(y_0))^2\big] \leq C(L,M)\, n^{-2/3},$$

where the infimum ranges over all estimators $\tilde{f}(y_0)$ based on $(Y_1, \ldots, Y_n)$.

1.4 Global Estimation and Inference

Remark 1.4.1. Formally, $f = F'$. One might try to estimate $f(y_0)$ by differentiating the empirical CDF

$$\hat{F}_n(y) \coloneqq \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{Y_i \leq y\}.$$

But $\hat{F}_n$ is a step function with jumps at the order statistics; it is not differentiable in the usual sense, and its derivative is 0 wherever it exists. This is why we introduce smoothing (histograms/kernels/local polynomials).

Definition 1.4.2 (Integrated squared error / integrated MSE).

For an estimator $\hat{f}$ of the entire density f on $[0,1]$, define the (weighted) integrated risk

$$\mathscr{R}(\hat{f}, f) := \mathbb{E}\Big[\int_0^1 \big(\hat{f}(y) - f(y)\big)^2 w(y)\, dy\Big],$$

often with $w(y) \equiv 1$.

Remark 1.4.3. If the pointwise bound in Proposition 1.3.7 holds uniformly in y_0 (as it does here), minimizing over h (which entails equating the bias and variance) then integrating yields the same rate:

$$\mathbb{E}\Big[\int_0^1 \big(\hat{f}_h(y) - f(y)\big)^2 dy\Big] = \int_0^1 \mathbb{E}\big[(\hat{f}_h(y) - f(y))^2\big]\, dy \lesssim n^{-2/3}.$$

Definition 1.4.4 (Confidence interval).

A $(1-\alpha)$ confidence interval for $f(y_0)$ is a random interval $I(Y_1, \ldots, Y_n) = [\ell, r]$ such that

$$\mathbb{P}\big(f(y_0) \in I\big) \geq 1 - \alpha \qquad \text{for all } P \in \mathscr{P}(L, M).$$

Remark 1.4.5 (Classical normal-approximation heuristic). A familiar recipe is

$$I \approx \hat{\theta} \pm z_{1-\alpha/2}\, \mathrm{se}(\hat{\theta}), \qquad \mathrm{se}(\hat{\theta}) := \sqrt{\mathrm{Var}(\hat{\theta})},$$

justified by the heuristic

$$\frac{\hat{\theta} - \theta}{\mathrm{se}(\hat{\theta})} \approx N(0, 1).$$

In nonparametric problems this can fail because the bias is often not negligible at the "MSE-optimal" tuning.

Remark 1.4.6. When we write

$$\frac{\hat{\theta} - \theta}{\mathrm{se}(\hat{\theta})} = \frac{\hat{\theta} - \mathbb{E}[\hat{\theta}]}{\mathrm{se}(\hat{\theta})} + \frac{\mathbb{E}[\hat{\theta}] - \theta}{\mathrm{se}(\hat{\theta})},$$

the first term is the variance term (often $\approx N(0, 1)$ after normalization), while the second term is the bias divided by standard error. In many classical settings this second term is $o(1)$; in nonparametrics it can be order 1 if h is chosen to balance bias and variance.

In other words, in parametric estimation, the estimator is often consistent (at least unbiased asymptotically), so the second term is zero and the statistic is indeed normally distributed. But we sometimes give up this quality in nonparametric estimation to reduce variance.

Remark 1.4.7 (Approach 1: undersmoothing). Choose h smaller than the MSE-optimal order so that bias is negligible relative to standard error. For the histogram, $\mathrm{se}(\hat{f}_h(y_0)) \asymp (nh)^{-1/2}$ and $\mathrm{Bias}(\hat{f}_h(y_0)) \lesssim h$, so a sufficient condition is

$$\frac{h}{(nh)^{-1/2}} = \sqrt{n}\, h^{3/2} \to 0, \qquad \text{i.e.} \qquad h = o(n^{-1/3}).$$

This makes the CI wider (more variance) but improves coverage by killing the bias term.

Remark 1.4.8 (Approach 2: bias correction). Estimate the bias and subtract it from the point estimator, then account for the increased variability. A modern "robust bias correction" viewpoint emphasizes that bias estimation inflates variance and the standard error must reflect that. (In RD, see Calonico–Cattaneo–Titiunik (2014) for a canonical development.)

Remark 1.4.9 (Approach 3: inference on a smoothed target). A different way to obtain valid inference is to change *what* we are doing inference for. Instead of targeting the point value $f(y_0)$, interpret the estimand as the *smoothed* (bin-averaged) quantity associated with the chosen resolution h. Indeed, for the histogram estimator one has

$$\mathbb{E}[\hat{f}_h(y_0)] = \frac{1}{h}\mathbb{P}(Y \in I_j) = \frac{1}{h}\int_{I_j} f(y)\,dy,$$

the average value of the density over the bin containing y_0 (hence a piecewise-constant approximation to f).

Viewed this way, $\hat{f}_h(y_0)$ is *unbiased* for the target $\frac{1}{h}\mathbb{P}(Y \in I_j)$, and standard large-sample (or binomial) arguments yield confidence intervals with correct coverage for this smoothed functional. The key point is conceptual: we are not claiming to learn $f(y_0)$ itself at the same level of certainty. Rather, we report uncertainty for a quantity that is directly identified by the data at the fixed smoothing level, so "coverage of the target" does not depend on controlling the smoothing bias relative to $f(y_0)$.

Aside 1.4.10. A related (and more "honest") variant is: upper bound the bias using smoothness (e.g. Lipschitz) and inflate the CI by that bound. This produces valid coverage at the cost of wider intervals.

Introduction to Nonparametric Regression 2

2.1 Framework

Notation 2.1.1.

Let $Z_i := (X_i, Y_i)$ with $Z_i \overset{\text{iid}}{\sim} P$ on $\mathbb{R}^d \times \mathbb{R}$. Write $X_i \in \mathbb{R}^d$ and $Y_i \in \mathbb{R}$. Given covariates X, we want to predict Y.

Definition 2.1.2 (Squared-error (prediction) risk).

For any measurable regression function $m : \mathbb{R}^d \to \mathbb{R}$, define its (population) squared-error risk

$$R_P(m) := \mathbb{E}_P\big[(Y - m(X))^2\big].$$

Definition 2.1.3 (Regression function / CEF).

The *regression function* (a.k.a. the *conditional expectation function* (CEF)) is

$$m^*(x) := \mathbb{E}_P[Y \mid X = x].$$

Suppose we get a fresh draw (X, Y) and only observe Y. How should we choose $h(x)$ as an estimator of $m^*(x)$?

Proposition 2.1.4 (Optimality of the regression function).

For any measurable $m : \mathbb{R}^d \to \mathbb{R}$,

$$R_P(m) = R_P(m^*) + \mathbb{E}_P\big[(m^*(X) - m(X))^2\big] \quad \geq \quad R_P(m^*).$$

In particular, m^* minimizes the squared error risk $R_P(m)$ over all measurable m.

Proof of Proposition 2.1.4. Step 1 (add–subtract $m^*(X)$). Write

$$Y - m(X) = \big(Y - m^*(X)\big) + \big(m^*(X) - m(X)\big).$$

Squaring and taking $\mathbb{E}_P$ gives

$$\begin{aligned}\mathbb{E}_P\big[(Y - m(X))^2\big] = \mathbb{E}_P\big[(Y - m^*(X))^2\big] + \mathbb{E}_P\big[(m^*(X) - m(X))^2\big] \\ + 2\,\mathbb{E}_P\big[(Y - m^*(X))(m^*(X) - m(X))\big].\end{aligned}$$

Step 2 (the cross term is 0). Using iterated expectations,

$$\mathbb{E}_P\big[(Y - m^*(X))(m^*(X) - m(X))\big] = 0,$$

because $\mathbb{E}_P\big[Y - m^*(X) \mid X\big] = 0$.

Step 3 (conclude). Substituting the vanishing cross term yields

$$R_P(m) = R_P(m^*) + \mathbb{E}_P\big[(m^*(X) - m(X))^2\big] \geq R_P(m^*),$$

as claimed.

Remark 2.1.5. The identity

$$R_P(m) - R_P(m^*) = \mathbb{E}_P\big[(m(X) - m^*(X))^2\big]$$

is a Pythagorean decomposition in $L^2(P)$: m^* is the L^2 projection of Y onto measurable functions of X.

Definition 2.1.6 (Excess risk).

Given an estimator $\hat{m}$ constructed from data, a natural target is the *excess risk* (or $L^2(P_X)$ error)

$$\mathbb{E}_P\big[(\hat{m}(X) - m^*(X))^2\big],$$

often interpreted as mean-squared prediction error with respect to the marginal distribution of X.

Given a finite training set $D_{\text{train}} = \{(X_i, Y_i)\}$, how do we estimate the true function that governs $y = h(x)$?

One approach is to use a regressogram: a piecewise constant function that is analogous to a histogram. Alternatively, we can make explicit modeling assumptions, such as only considering

These are parametric models.

$$m(x) = \sum_{j=1}^{K} \beta_j \psi_j(x), \text{ or } m(X) = m(x_j, \beta).$$

We can also use supervised learning like deep learning, trees, random forests, and boosting.

Nonparametric regression is thus a problem we're very good at solving. One reason is that we can see how well we are doing, but not really *why* we are good. One topic in nonparametric regression today is repurposing existing black boxes for new statistical tasks.

Remark 2.1.7. Suppose $X_i \in \{1, \ldots, K\}$ is a categorical random variable. We might estimate by

$$\hat{m}(k) = \frac{1}{\#\{X_i = K\}} \sum_{i : X_i = k} Y_i.$$

This estimator is only defined when every bin (category) is attained by some observation in the training data. If there is k' where no $X_i = k'$, we might put $\hat{m}$ as 0 or as the global average of Y.

We would expect $\text{Bias}(\hat{m}(k)) \approx 0$ and $\text{Var}(\hat{m}(k)) \approx \frac{\sigma^2}{\#\{X_i = k\}}$. The first approximation is because if $\#\{X_i = k\} = 0$, we put $\hat{m}(k) = 0$, which introduces bias if $m(k) \neq 0$. The second approximation is because $\text{Var}(Y_i \mid X = x) \leq \sigma^2$ and $\#\{X_i = k\}$ itself is random.[1]

One resolution to the second approximation is a conditional bias-variance decomposition. So far, we have been examining

$$\mathbb{E}\left[(\hat{m}(x_0) - m(x_0))^2\right] = \text{Bias}^2 + \text{Var}.$$

Instead, we could study the conditional decomposition

$$\mathbb{E}\left[(\hat{m}(x_0) - m(x_0))^2 \mid X_{1:n}\right] = \text{Var}(\hat{m}(x_0) \mid X_{1:n}) + \left(\mathbb{E}[\hat{m}(x_0) \mid X_{1:n}] - m(x_0)\right)^2.$$

We have two difference goals. There are two distinct targets one might mean by "the regression function" once we discretize X by creating bins.

(1) **Discrete-design goal (exact categories).** Suppose X is actually discrete and we want

$$m(k) := \mathbb{E}[Y \mid X = k].$$

With the cell-mean estimator

$$\hat{m}(k) := \frac{1}{N_k} \sum_{i:X_i = k} Y_i, \qquad N_k := \#\{i : X_i = k\},$$

we have, whenever $N_k > 0$,

$$\mathbb{E}[\hat{m}(k) \mid X_{1:n}] = \frac{1}{N_k} \sum_{i:X_i = k} \mathbb{E}[Y_i \mid X_{1:n}] = \frac{1}{N_k} \sum_{i:X_i = k} m(k) = m(k),$$

so the conditional bias relative to the target $m(k)$ is 0. In this setting, the only issue is what to do when $N_k = 0$. We might define $\hat{m}(k) := 0$ in this case, so the bias would be $m(k)$.

(2) **Regressogram goal (binning a continuous regressor).** Now take X continuous and partition the support into bins $(I_j)_j$. For $x_0 \in I_j$, define the regressogram

$$\hat{m}(x_0) := \frac{1}{N_j} \sum_{i:X_i \in I_j} Y_i, \qquad N_j := \#\{i : X_i \in I_j\}.$$

Even if $N_j > 0$,

$$\mathbb{E}[\hat{m}(x_0) \mid X_{1:n}] = \frac{1}{N_j} \sum_{i:X_i \in I_j} \mathbb{E}[Y_i \mid X_{1:n}] = \frac{1}{N_j} \sum_{i:X_i \in I_j} m(X_i),$$

[1] Recall the law of total expectation; we have that $\mathbb{E}[\text{Var}(\hat{m}(k))] \approx \frac{\sigma^2}{\#\{X_i = k\}}$, but $\text{Var}(\mathbb{E}[\hat{m}(k)]) > 0$.

so the conditional bias relative to the *pointwise* target $m(x_0)$ is

$$\mathbb{E}\left[\hat{m}(x_0) \mid X_{1:n}\right] - m(x_0) = \frac{1}{N_j} \sum_{i: X_i \in I_j} \left(m(X_i) - m(x_0)\right),$$

which is generally nonzero unless $m(\cdot)$ is constant on I_j or all $X_i = x_0$.

A natural alternative target is the *bin mean*

$$m_j \coloneqq \mathbb{E}\left[Y \mid X \in I_j\right] = \mathbb{E}\left[m(X) \mid X \in I_j\right],$$

for which the regressogram is (conditionally) unbiased when $N_j > 0$. But if the goal is $m(x_0)$, then there is an unavoidable discretization/smoothing bias.

This is why we assume $m(\cdot)$ is L-Lipschitz: if $x_0 \in I_j$ and the bin half-width is h, then for $X_i \in I_j$,

$$|m(X_i) - m(x_0)| \leq L\, |X_i - x_0| \leq Lh,$$

hence

$$\left|\mathbb{E}\left[\hat{m}(x_0) \mid X_{1:n}\right] - m(x_0)\right| \leq Lh.$$

We can then recover the unconditional MSE decomposition:

$$\begin{aligned}\mathbb{E}\left[\left(\hat{m}(x_0) - m(x_0)\right)^2\right] &= \mathbb{E}\left[\mathbb{E}\left[\left(\hat{m}(x_0) - m(x_0)\right)^2 \mid X_{1:n}\right]\right] \\ &= \mathbb{E}\left[\operatorname{Var}\left(\hat{m}(x_0) \mid X_{1:n}\right)\right] + \mathbb{E}\left[\operatorname{Bias}^2\left(\hat{m}(x_0) \mid X_{1:n}\right)\right].\end{aligned}$$

2.2 k-Nearest Neighbors Estimator

Suppose we want to estimate $m(x_0)$.

Definition 2.2.1 (k-nearest neighbors of a point).

The k-nearest neighbors of a point x_0 are

$$N_k(x_0) = \{i \in \{1, \ldots, \} : \|X_i - x_0\| \text{ is among the } k \text{ smallest}\}.$$

If there are ties in distances, this is not well-defined. For simplicity, we will assume there are no ties.

Definition 2.2.2 (k-nearest neighbors estimator).

Define the kNN estimator as

$$\hat{m}(x_0) = \frac{1}{k} \sum_{i \in N_k(x_0)} Y_i.$$

Consider the 1-nearest neighbor estimator. We have $\hat{m}(X_i) = Y_i$, and the MSE is $\sum \left(Y_i - \hat{m}(X_i)\right)^2 = 0$. The traditional view is that this is major overfitting and thus bad, but the modern view is that it is not necessarily bad.

Note the MSE here is being evaluated at points where we actually have an observation x_i.

We may have *boundary bias*, which is important in something like regression discontinuity design. This arises because our k neighbors at the boundary are evenly distributed around the estimated point, which causes a problem if we have a trend near the boundary.

Bias increases as k increases, while variance is decreasing in k. We will now formalize this.

Proposition 2.2.3 (Rate of MSE of k-NN estimator).

Assuming the following:

(i) $\operatorname{Var}(Y_i \mid X = x) \leq \sigma^2$ for all x

(ii) $m(\cdot)$ is L-Lipschitz

(iii) $X_i \sim U\,[0,1]^d$,

the rate of MSE decay of the k-nearest neighbors estimator as a function of n with optimal[a] $k^* \asymp n^{\frac{2}{2+d}}$ is given by

$$\mathbb{E}\left[\left(\hat{m}_{k^*}(x_0) - m(x_0)\right)^2\right] \lesssim n^{-\frac{2}{2+d}}.$$

For arbitrary k,

$$\mathbb{E}\left[\left(\hat{m}_k(x_0) - m(x_0)\right)^2\right] \lesssim_{\sigma^2, L} \frac{\sigma^2}{k} + L^2\left(\frac{k}{n}\right)^{\frac{2}{d}}.$$

[a] In the minmax sense with these minimal assumptions.

Proof of Proposition 2.2.3. We will use the conditional MSE. Start with

There is randomness due to the *design* (location of X_is) and randomness of Y_i once we fix X_i. Starting by conditioning on X_i helps us treat these sources one at a time.

$$\begin{aligned}
\left|\operatorname{Bias}\left(\hat{m}(x_0) \mid X_{1:n}\right)\right| &= \left|\mathbb{E}\left[\hat{m}(x_0) \mid X_{1:n}\right] - m(x_0)\right| \\
&= \left|\mathbb{E}\left[\frac{1}{k}\sum_{i \in N_k(x_0)} Y_i \mid X_{1:n}\right] - m(x_0)\right| \\
&= \frac{1}{k}\left|\sum_{i \in N_k(x)} \underbrace{\mathbb{E}\left[Y_i \mid X_{1:n}\right]}_{=\mathbb{E}[Y_i \mid X_i] = m(X_i)} - m(x_0)\right| \\
&= \frac{1}{k}\left|\sum_{i \in N_k(x_0)} m(X_i) - m(x_0)\right| \\
&\leq \frac{1}{k}\sum_{i \in N_k(x_0)} \left|m(X_i) - m(x_0)\right| \\
&\leq L\|X_i - x_0\| \\
&\leq L\|X_{(k)}(x_0) - x_0\|.
\end{aligned}$$

$$\begin{aligned}
\operatorname{Var}\left(\hat{m}(x_0) \mid X_{1:n}\right) &= \operatorname{Var}\left(\frac{1}{k} \sum_{i \in N_k(x_0)} Y_i \mid X_{1:n}\right) \\
&= \frac{1}{k^2} \sum_{i \in N_k(x_0)} \operatorname{Var}\left(Y_i \mid X_i\right) \\
&= \frac{\sigma^2}{k}.
\end{aligned}$$

Hence the MSE is bounded by

$$\begin{aligned}
\mathrm{MSE} &= \mathbb{E}\left[\left(\hat{m}(x_0) - m(x_0)\right)^2\right] && (2.1) \\
&\leq \frac{\sigma^2}{k} + L^2 \mathbb{E}\left[\|X_{(k)}(x_0) - x_0\|^2\right]. && (2.2)
\end{aligned}$$

Note that the bound on unconditional bias is random because the bound on conditional bias depends on the covariates — we must take an expectation.

Now using assumption (iii) with $d = 1$, we expect

$$\mathbb{E}\left[\|X_{(k)}(x_0) - x_0\|^2\right] \approx \left(\frac{k}{n}\right)^2.$$

In d dimensions, this is less interpretable[2]:

As dimension grows, the k nearest neighbor will be much further away. We run into the curse of dimensionality again.

$$\mathbb{E}\left[\|X_{(k)}(x_0) - x_0\|^2\right] \approx \left(\frac{k}{n}\right)^{\frac{2}{d}}.$$

We substitute this into our MSE bound to get

$$\mathrm{MSE} \lesssim_{L,\sigma^2} \frac{\sigma^2}{k} + L^2 \left(\frac{k}{n}\right)^{\frac{2}{d}}.$$

Refer to the lecture slides for some plots for different k. How do we actually pick the best choice of k? We find this by equating the variance and square bias, up to the removal of constants (because we just want to derive a $O(\cdot)$ bound):

Another way to actually pick the optimal k without making an assumption on the actual form of m is called cross-validation; we will learn about this later in the course!

$$\begin{aligned}
\frac{1}{k} &= \left(\frac{k}{n}\right)^{\frac{2}{d}} \\
k^{\frac{2+d}{d}} &= n^{\frac{2}{d}} \\
k^* &= n^{\frac{2}{2+d}}.
\end{aligned}$$

For $d = 1$, we see $k^* = n^{\frac{2}{3}}$ and for $d = 2$, we get $k^* = n^{\frac{1}{2}}$. As the dimenion grows, the number of neighbors we select shrinks due to the bias blowing up.

Plugging this k^* into our bound on the MSE, we get that the rate of MSE growth as a function of n is given by

$$\mathrm{MSE} \lesssim n^{-\frac{2}{2+d}}.$$

For $d = 1$,

$$\mathrm{MSE} \lesssim n^{-\frac{2}{3}}.$$

Recall that for a Lipschitz density, the minimax MSE was also

$$\inf_{m(\cdot)\ \text{L-}Lipschitz} \gtrsim n^{-\frac{2}{2+d}}.$$

Proof. (Of d-dimension distance bound.) Fix $x_0 \in [0, 1]$. We know the bound that

[2]We provide the proof for this distance bound after this proof concludes.

for any ϵ, there exists some c_d such that

$$\mathbb{P}\left[\|X_{(i)} - x_0\| \le \epsilon\right] \ge c_d \epsilon^d.$$

This constant exists because of our assumption that $X_i \sim \mathrm{Unif}[0,1]^d$.

We can write (being lax with constants changing)

$$\begin{aligned}
\mathbb{E}\left[\|X_{(1)}(x_0) - x_0\|^2\right] &= \int_0^\infty \mathbb{P}\left[\|X_{(1)}(x_0) - x_0\|^2 \ge \epsilon\right] d\epsilon \\
&= \int_0^\infty \mathbb{P}\left[\cap_i \{\|X_i - x_0\| \ge \sqrt{\epsilon}\}\right] d\epsilon \\
&= \int_0^\infty \prod_{i=1}^n \mathbb{P}\left[\|X_i - x_0\| \ge \sqrt{\epsilon}\right] d\epsilon \\
&\le \int_0^\infty \left(1 - c_d \epsilon^{\frac{d}{2}}\right)^n d\epsilon \\
&\le \int_0^\infty \exp\left(-n c_d \epsilon^{\frac{d}{2}}\right) d\epsilon \\
&= \int_0^\infty n^{-\frac{2}{d}} c_d^{-\frac{2}{d}} \exp\left(-u^{\frac{d}{2}}\right) du \\
&\le c \left(\frac{1}{n}\right)^{\frac{2}{d}}.
\end{aligned}$$

Now we upgrade from 1-NN to K-NN. Suppose n is a multiple of $2K$ and partition $\{1, \dots, n\}$ into $2K$ equal blocks $I_1, \dots, I_{2K}$.

Write $X_{(k)}(x_0; I)$ for the kth nearest neighbor of x_0 among $\{X_i : i \in I\}$, and set

$$R_k \coloneqq \left\|X_{(k)}(x_0; \{1, \dots, n\}) - x_0\right\|, \qquad R_{1,j} \coloneqq \left\|X_{(1)}(x_0; I_j) - x_0\right\|$$

for blocks $I_1, \dots, I_{2k}$ forming a partition of $\{1, \dots, n\}$ (so $|I_j| \approx n/(2k)$). We claim $R_{1,j} \ge R_k$ for at least k blocks $j \in \{1, \dots, 2k\}$. If $R_{1,j} < R_k$ held in k distinct blocks, then in each such block one could pick an index $i_j \in I_j$ with $\|X_{i_j} - x_0\| < R_k$. These k indices are distinct (because the blocks are disjoint), so there would be at least k sample points strictly closer than R_k to x_0, contradicting the definition of R_k as the kth order statistic of the distances $\{\|X_i - x_0\|\}_{i=1}^n$.

Consequently, at least k terms in the sum $\sum_{j=1}^{2k} R_{1,j}^2$ are $\ge R_k^2$, so

$$\sum_{j=1}^{2k} R_{1,j}^2 \ge k R_k^2, \qquad \text{and hence} \qquad R_k^2 \le \frac{1}{k} \sum_{j=1}^{2k} R_{1,j}^2.$$

Taking expectations and using linearity,

$$\begin{aligned}
\mathbb{E}\left[R_k^2\right] &\le \frac{1}{k} \sum_{j=1}^{2k} \mathbb{E}\left[R_{1,j}^2\right] \\
&= \frac{2k}{k} \mathbb{E}\left[R_{1,1}^2\right] \qquad \text{(each block has the same distribution).}
\end{aligned}$$

Plug in the 1-NN bound at the block size. Let $m \coloneqq |I_1|$ (so $m \asymp n/(2k)$). We can use the 1-NN bound at this block size to get

$$\mathbb{E}\left[R_{1,1}^2\right] \le C\, m^{-2/d}.$$

Therefore,

$$\mathbb{E}\left[R_k^2\right] \leq 2C\, m^{-2/d} \leq 2C\left(\frac{2k}{n}\right)^{2/d} \leq C'\left(\frac{k}{n}\right)^{2/d},$$

□

Remark 2.2.4. Suppose we have n observations with a given MSE m, written as (n, m). What n' do we need for a new MSE $m' = \frac{m}{2}$? If the rate is $\frac{1}{n}$, then we require $n' = 2n$. If the rate is instead $\frac{1}{\sqrt{n}}$, we require $n' = 4n$.

What are other assumptions we could make to give faster convergence than in (2.2)? In order, we will explore:

(i) Additional smoothness assumptions (Hölder classes)

(ii) Structural assumption

Definition 2.2.5 (Hölder class).

The $\Sigma(\beta, L)$ Hölder class for $\beta > 0$ is defined such that $m \in \Sigma(\beta, L)$ if it is l times differentiable and

$$\left|m^{(l)}(x) - m^{(l)}(x')\right| \leq L\left|x - x'\right|^{\beta - l},$$

where $l := \lfloor \beta \rfloor$ is the largest integer strictly less than β. In particular, if $\beta \in \mathbb{N}$ then $l = \lfloor \beta \rfloor = \beta - 1$.

Remark 2.2.6. $\Sigma(1, L)$ is the class of L-Lipschitz functions and $\Sigma(2, L)$ are those with L-Lipschitz continuous derivatives.

Remark 2.2.7. Minimax rates are roughly $n^{-\frac{2\beta}{2\beta+d}}$. As we have larger β, we get a faster rate of convergence, as we can write $n^{-\frac{2s}{2s+1}}$ for $s = \frac{\beta}{d}$. As $s \to \infty$, the rate converges to $\frac{1}{n}$. However, in general, k-NN does not achieve this rate.

Now we consider structural assumptions. We can suppose an additive model, where

$$m(x) = \sum_{j=1}^{d} m_j(x_j)$$

for L-Lipschitz m_j. We can also assume $m(\cdot)$ looks like a neural network.

Imposing shape contrails like monotonicity, convexity, log-concavity. These are usually analyzed for losses such as

$$\mathbb{E}\left[(\hat{m}(X) - m(X))^2\right] = \int \mathbb{E}\left[(\hat{m}(x_0) - m(x_0))^2\right] f(x_0)dx_0,$$

where $X \sim \mathbb{P}$.

This is sometimes called the integrated MSE or the expected test error. In 1-D, we can achieve the same rate of $n^{-\frac{2}{3}}$ if we replace the smoothness assumption with a monotonicity assumption on m.

Definition 2.2.8 (Manifold hypothesis).

Suppose in d dimensions, our data has "effective dimension"[a] $s < d$. Kpotufe (2011) showed that k-NN with a Lipschitz assumption achieves the rate $n^{-\frac{2}{2+s}}$.

[a]Think of a curve in R^2, which has effective dimension 1.

What if we relax assumption (iii) in that we now just assume X_i has a density f such that $f(x) \geq \delta > 0$ for all $x \in [0,1]^d$? In our bound on the d-dimension distance, we must use

$$\mathbb{P}\left[\|X_i - x_0\| \leq \epsilon\right] \geq c_d \delta \epsilon^d.$$

For a fixed δ, the rate is still the same. However, the MSE depends on δ.

If we relax assumption (iii) by even allow $\inf_{x_0} f(x_0) = 0$? We cannot port this into the analysis we have already done; we evaluate integrated MSE.

Aside 2.2.9 (Integrated MSE). Examining

$$\mathbb{E}\left[(\hat{m}(X) - m(X))^2\right] = \int \mathbb{E}\left[(\hat{m}(x_0) - m(x_0))^2\right] f(x_0) dx_0,$$

we justified that low density means few nearby neighbors and potentially large local MSE. High density means the same local MSE may be smaller.

We have the following theorem:

Suppose $X_i \in [0,1]^d$, $\operatorname{Var}(Y_i \mid X_i = x) \leq \sigma^2$ for all x, and $m(\cdot)$ is L-Lipschitz. For optimally tuned k-NN, we have the bound

This is the same result as before, except now we do not assume $X \sim U[0,1]^d$.

$$\mathbb{E}\left[(\hat{m}(X) - m(X))^2\right] \lesssim_{\sigma^2, L} n^{-\frac{2}{2+d}}$$

We have no assumption on X's density.

for $d \geq 3$.

Theorem 2.2.10 (Universal consistency of k-NN estimator).

Suppose $(X_i, Y_i) \overset{iid}{\sim} \mathbb{P}$, $\mathbb{E}\left[Y_i^2\right] < \infty$, and k is such that $\frac{k}{n} \to 0$ as $k \to \infty$. Then for the k-NN estimator $\hat{m}_k$,

$$\mathbb{E}\left[(\hat{m}_k(X) - m(X))^2\right] \to_{n \to \infty} 0.$$

This result is notable because there is no smoothness assumption on the true function m. The price we pay for this is no guaranteed rate of convergence.

Kernel Methods 3

3.1 Kernel Density Estimation

Definition 3.1.1 (Regressogram).

The regressogram is an estimator of the conditional expectation of a random variable and is given by

$$\hat{m}(x;h) = \begin{cases} \frac{\sum_{i=1}^{n} \mathbb{1}_{X_i \in B_k} Y_i}{\sum_{i-1}^{n} \mathbb{1}_{X_i \in B_k}}, & x \in B_k \end{cases},$$

where h is a parameter that determines the width of each bin B_k.

Table 3.1: k-NN and Regressogram Comparison

	Advantages	Disadvantages
k-NN	• Closer to continuous (?)	• Discontinuous • Boundary bias • Storage (need full dataset)
Regressogram	• Storage: only need boundaries + function values	• Discontinuous • Boundary bias within bins • Choosing bin boundaries and widths • There may be no training points in the bin of x_0

While the regressogram has boundary bias within the bins, we can solve this with an "sliding window" estimator that combines k-NN and the regressogram.

Definition 3.1.2 (Local average estimator).

Define the local average estimator

$$\hat{m}(x_0) = \frac{\sum_{i:|X_i - x_0| \le h} Y_i}{\sum_i \mathbb{1}_{|X_i - x_0| \le h}} = \frac{\sum_{i:|X_i - x_0| < h} Y_i}{\#\{i : |X_i - x_0| \le h\}}.$$

There may be no points within h of x_0, and the local average estimator is still not continuous.

Definition 3.1.3 (Nadaraya-Watson).

Given $K(\cdot)$ and h, define the Nadaraya-Watson estimator (local weighted average) as

$$\hat{m}(x_0) = \frac{\sum_{i=1}^{n} K\left(\frac{X_i - x_0}{h}\right) \cdot Y_i}{\sum_{i=1}^{n} K\left(\frac{X_i - x_0}{h}\right)}.$$

We can get the local average estimator by putting $K(u) = \frac{1}{2}\mathbb{1}_{|u|\leq 1}$.

The $\frac{1}{2}$ arises from normalizing K to be a density. Note that this factor cancels out.

Example 3.1.4.

One popular kernel is the Epanechnikov Kernel, which puts

$$K(u) \propto \left(1 - u^2\right) \mathbb{1}_{|u|\leq 1}.$$

Another is the Gaussian kernel, where

$$K(u) \propto \exp\left(-\frac{u^2}{2}\right).$$

We also have the tri-cube kernel:

$$K(u) \propto (1 - u^3)\mathbb{1}_{|u|\leq 1}$$

and the triangular kernel:

$$K(u) \propto (1 - |u|)\, \mathbb{1}_{(|u|\leq 1)}.$$

We can think of these kernels as density functions, noting that they are symmetric around 0 and oftentimes have compact support of $[-1, 1]$.

Remark 3.1.5. If $K(\cdot)$ is continuous and the denominator is positive, then $\hat{m}$ is continuous. There is still boundary bias, but it is slightly reduced (since the points further from the boundary are weighted less).

Before studying Nadaraya-Watson, we want to study the analogous idea for density estimation, forming a local weighted histogram estimator. Suppose $X_i \sim f(\cdot)$ and $\mathrm{Supp}(f) \subseteq [0, 1]$.

Definition 3.1.6 (Kernel density estimator).

Define the kernel density estimator (local weighted histogram estimator) as

$$\hat{f}(x_0) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{X_i - x_0}{h}\right).$$

Remark 3.1.7. If K is the Gaussian kernel, then

$$\hat{f}(x_0) = \frac{1}{n}\sum_{i=1}^{n} N(X_i - x_0, h^2).$$

Then $\hat{f}$ is the sum of n normal RV densities.

Proposition 3.1.8.

Assume that

- $\text{Supp}(K) = [-1, 1]$, K nonnegative
- $\int_{-1}^{1} K(u)du = 1$
- $x_0 \in [0, 1]^\circ$, so we are guaranteed $[x_0 - h, x_0 + h] \subseteq [0, 1]$ for some h.
- $f(\cdot)$ is continuous at x_0.

Then $\hat{f}(x_0)$ is consistent.

Proof of Proposition 3.1.8. We can write

It may also be required that K is square integrable

This kind of change-of-variables is common with NW-type estimators.

$$\begin{aligned}
\mathbb{E}\left[\hat{f}(x_0)\right] &= \frac{1}{h}\mathbb{E}\left[K\left(\frac{X_i - x_0}{h}\right)\right] \\
&= \frac{1}{h}\int_0^1 K\left(\frac{z - x_0}{h}\right) f(z)dz \\
&= \frac{1}{h}h\int_{-\frac{x_0}{h}}^{\frac{1-x_0}{h}} K(u)f(x_0 + uh)du, \quad u = \frac{z - x_0}{h} \\
&= \int_{-1}^{1} K(u)f(x_0 + uh)du \\
&\overset{h\downarrow 0}{\to} \int_{-1}^{1} K(u)f(x_0)du \quad \text{by continuity of } f \text{ at } x_0 \\
&= f(x_0)\int_{-1}^{1} K(u)du \\
&= f(x_0).
\end{aligned}$$

We see that the bias goes to 0 as $h \to 0$. Similar to the argument we used for the unweighted histogram estimator, we can show that $\mathrm{Var}\left(\hat{f}(x_0)\right) \lesssim \frac{1}{nh} \xrightarrow{nh\to\infty} 0$.

We put the limits of integration as -1 and 1 because of our assumption on the support of K. Due to the restriction on the support of K, we get boundary bias. If $x_0 = 0$, then the lower limit of integration is 0 instead of -1, and we end with $\mathbb{E}\left[\hat{f}(x_0)\right] = \frac{1}{2} f(x_0)$.

We have shown consistency in the interior.

In fact, at the boundary, the kernel density estimator satisfies

$$\hat{f}(0) \xrightarrow{p} \frac{1}{2} f(0).$$

Hence, the KDE is inconsistent at the boundary. However, kernel regression with the Nadaraya-Watson estimator is consistent even at the boundary, but with slower convergence.

Once we have obtained a KDE $\hat{f}$, how can we sample a random variable according to this distribution?

First, consider the ECDF $\hat{F}$. We can sample $I \sim \mathrm{Unif}(1, \dots, n)$, then sample $X \sim \mathrm{Unif}\{X_1, \dots, X_n\}$. We only sample points that we have observed; the dataset is the distribution.

The set X is sampled from allows for repetitions.

While we could sample $U \sim \mathrm{Unif}(0, 1)$, integrate $\hat{f}$ to derive $\hat{F}$, invert to get $\hat{F}^{-1}$, and set $X = \hat{F}^{-1}(U)$, we see that as $h \to 0$, the integral of the KDE approximates the ECDF, as it will approach placing mass $\frac{1}{n}$ at each sample point X_i.

To sample from a KDE, recall the KDE with bandwidth $h > 0$,

$$\hat{f}_h(x) \coloneqq \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{X_i - x_0}{h}\right).$$

This is a finite mixture of the n kernel components centered at the sample points. In particular, let

$$I \sim \mathrm{Unif}\left(\{1, \dots, n\}\right), \qquad \epsilon \text{ with density } K,$$

be independent, and define

$$X \coloneqq X_I + h\epsilon.$$

Then, conditional on $I = i$, X has density

$$f_{X|I=i}(x_0) = \frac{1}{h} K\left(\frac{X_i - x_0}{h}\right),$$

so by averaging over I,

$$f_X(x_0) = \frac{1}{n} \sum_{i=1}^{n} f_{X|I=i}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{X_i - x_0}{h}\right) = \hat{f}_h(x).$$

Essentially, the KDE $\hat{f}_h$ is the density of the mixture

$$X = X_I + h\epsilon,$$

where $I \sim \mathrm{Unif}\left(\{1, \dots, n\}\right)$ and ϵ has density K (independent of I). We first pick an index I uniformly, then add "kernel noise" $h\epsilon$ around X_I. Assuming that K peaks around 0, we are more likely to pick near observations X_i than far from them.

For the Gaussian kernel $K(u) = \phi(u)$, this means $\epsilon \sim N(0,1)$ and

$$X \mid I = i \sim N\left(X_i, h^2\right).$$

As $h \to 0$, the added noise $h\epsilon$ vanishes, so X converges in distribution to the empirical distribution supported on $\{X_1, \ldots, X_n\}$.

3.2 Nadaraya-Watson Estimator

Recall that the Nadaraya-Watson estimator is defined as

$$\hat{m}(x_0) = \frac{\sum_{i=1}^{n} K\left(\frac{X_i - x_0}{h}\right) \cdot Y_i}{\sum_{i=1}^{m} K\left(\frac{X_i - x_0}{h}\right)}.$$

We have the following standing assumptions:

(i) $m(\cdot)$ is twice continuously differentiable at x_0

(ii) $\sigma^2(x) \coloneqq \operatorname{Var}\left(Y_i \mid X_i = x\right)$ is continuous at x_0 and $\sigma^2(x_0) > 0$.

(iii) $\mathbb{P}_X$ is continuously differentiable at x_0, and $f(x_0) > 0$

(iv) $x_0 \in [0,1]^\circ$

(v) $\operatorname{Supp}(K(\cdot)) = [-1,1]$ and

$$\begin{aligned} \int_{-1}^{1} K(u)du &= 1, \\ \int_{-1}^{1} K^2(u)du &< \infty, \\ \int_{-1}^{1} uK(u)du &= 0, \\ \int_{-1}^{1} u^2K(u)du &< \infty \end{aligned}.$$

We no longer assume that K is non-negative.

Define $\epsilon_i = Y_i - m(X_i)$. Then

$$\begin{aligned} \hat{m}(x_0) - m(x_0) &= \frac{\sum_{i=1}^{n} K\left(\frac{X_i - x_0}{h}\right)\epsilon_i}{\sum_{i=1}^{n} K\left(\frac{X_i - x_0}{h}\right)} \\ &+ \frac{\sum_{i=1}^{n} K\left(\frac{X_i - x_0}{h}\right)\left(m(X_i) - m(x_0)\right)}{\sum_{i=1}^{n} K\left(\frac{X_i - x_0}{h}\right)} \end{aligned}.$$

The first term contributes to conditional variance and the second to conditional bias.[1]

[1] Conditioning on $X_1, \ldots, X_n$.

We will analyze $\operatorname{MSE}(\hat{m}(x_0) \mid X_{1:n})$ as $h \to 0$ and $nh \to \infty$. Eventually, we want some result of the form

$$\operatorname{MSE}(\hat{m}(x_0) \mid X_{1:n}) = \psi(n, h, K, \mathbb{P})(1 + o_{\mathbb{P}}(1)).$$

Here $\mathbb{P}$ enters through the assumptions on σ^2 and continuity. This is a pointwise-in-$\mathbb{P}$ analysis, not a uniform supremum over $\mathscr{P}$.

3.3 Interior and Boundary Asymptotics

We now analyze the pointwise MSE of the Nadaraya-Watson estimator, examining both interior and boundary behavior. The analysis proceeds in three steps:

1. **Variance bound:** Derive the conditional variance of $\hat{m}(x_0)$ given $X_{1:n}$.
2. **Bias bound:** Characterize the conditional bias using Taylor expansions.
3. **Comparison:** Compare interior vs. boundary rates and discuss optimal bandwidth selection.

Bounding the conditional variance. Conditioning on $X_{1:n}$,

$$\begin{aligned}
\operatorname{Var}(\hat{m}(x_0) \mid X_{1:n}) &= \operatorname{Var}\left(\frac{\sum K\left(\frac{X_i - x_0}{h}\right) Y_i}{\sum K\left(\frac{X_i - x_0}{h}\right)} \mid X_{1:n}\right) \\
&= \frac{1}{\left(\sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right)\right)^2} \sum_{i=1}^n K^2\left(\frac{X_i - x_0}{h}\right) \sigma^2(X_i) \\
&= \frac{\left(\frac{1}{nh}\right)^2}{\left(\underbrace{\frac{1}{nh}\sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right)}_{\xrightarrow{p} f(x_0)}\right)^2} \sum_{i=1}^n K^2\left(\frac{X_i - x_0}{h}\right) \sigma^2(X_i).
\end{aligned}$$

Looking at

$$\begin{aligned}
N &:= \frac{1}{nh} \sum_{i=1}^n K^2\left(\frac{X_i - x_0}{h} \sigma^2(X_i)\right) \\
\mathbb{E}[N] &= \frac{1}{h}\left(\mathbb{E}\left[K^2\left(\frac{X_i - x_0}{h}\sigma^2(X_i)\right)\right]\right) \\
&= \frac{1}{h}\int_0^1 K^2\left(\frac{z - x_0}{h}\right)\sigma^2(z) f(z)\,dz \\
&= \int_{-1}^1 K^2(u)\sigma^2(uh + x_0) f(uh + x_0)\,du, \quad u = \frac{z - x_0}{h} \\
&\xrightarrow{h \to 0} \int_{-1}^1 K^2(u)\sigma^2(x_0) f(x_0)\,du \\
&= f(x_0)\sigma^2(x_0)\int_{-1}^1 K^2(u)\,du,
\end{aligned}$$

Note that N is actually the numerator divided by $\frac{1}{nh}$; we correct for this later.

where the convergence follows from our continuity assumptions and u being fixed in the integrand. (We also used the assumption on the support of K.)

We can check $\operatorname{Var}(N) \to 0$, so $N \to f(x_0)\sigma^2(x_0)\int_{-1}^1 K^2(u)du$.

Substituting,

$$\operatorname{Var}(\hat{m}(x_0) \mid X_{1:n}) = \left[\frac{1}{nh}\frac{\sigma^2(x_0)}{f(x_0)}\int_{-1}^1 K^2(u)du\right](1 + o_{\mathbb{P}}(1)).$$

We earlier derived that $\operatorname{Var}(\hat{m}(x_0)) \lesssim \frac{1}{nh}$; now we have the constant. It makes sense for the conditional variance of Y_i at $X_i = x_0$ to be the numerator and $f(x_0)$ to be in the denominator.

If we assumed $x_0 = 0$, i.e., we are not in the interior, we get that the variance is double the interior variance.[2]

The expressions before are true only when h is small enough to guarantee the support of K in X_i is within the support of X_i.

Bounding the conditional bias. Similarly,

Practice this change-of-variables; it is common in bounds on kernel estimator MSEs.

$$\begin{aligned}
\operatorname{Bias}(\hat{m}(x) \mid X_{1:n}) &= \frac{\sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right)(m(X_i) - m(x_0))}{\sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right)} \\
&= \frac{\frac{1}{nh}\sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right)(m(X_i) - m(x_0))}{\underbrace{\frac{1}{nh}\sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right)}_{\xrightarrow{p} f(x_0)}} \\
\mathbb{E}[N'] &= \frac{1}{h}\int_0^1 K\left(\frac{z - x_0}{h}\right)(m(z) - m(x_0)) f(z)dz \\
&= \int_{-1}^1 K(u)(m(uh + x_0) - m(x_0)) f(uh + x_0)du \\
&\xrightarrow{h \to 0} \int_{-1}^1 K(u) \cdot 0 \cdot f(x_0)du = 0.
\end{aligned}$$

We can check $\operatorname{Var}(N') \to 0$, so $\operatorname{Bias}(\hat{m}(x_0) \mid X_{1:n}) \to 0 + o_{\mathbb{P}}(1)$ as $h \to 0$.

$o_{\mathbb{P}}(1)$ is a sequence of RVs Z_n such that $Z_n \xrightarrow{p} 0$.

Combined with our result on the conditional variance of $\hat{m}(x_0)$, we have that the NW estimator is consistent.

We can get a more precise result for the bias than just consistency, which says nothing about the speed of convergence. Write

$$\begin{aligned}
\mathbb{E}[N'] &= \int_{-1}^1 K(u)(m(uh + x_0) - m(x_0)) f(uh + x_0)du \\
&= h\int_{-1}^1 K(u)u \cdot m'(\xi_u) f(uh + x_0)du \text{ by diff. of } m. \\
&\xrightarrow{h \to 0} h\underbrace{m'(x_0) f(x_0)\int_{-1}^1 K(u)udu}_{\xrightarrow{p} f(x_0)m'(x_0)\int_{-1}^1 K(u)udu = 0},
\end{aligned}$$

[2] There's a factor of $\frac{1}{2}$ from the integral being from 0 to 1 and a factor of 2^2 from the change in the density.

so $N' = h \cdot o_{\mathbb{P}}(1)$. Since $D \xrightarrow{p} f(x_0)$, a constant,

$$\operatorname{Bias}\left(\hat{m}(x_0) \mid X_{1:n}\right) = \frac{N'}{D} = h \cdot o_{\mathbb{P}}(1),$$

which is a stronger result that the mere consistency we saw before.

If we had $x_0 = 0$ (on the boundary), our integral would be $\int_0^1 K(u)udu \neq 0$ and $f(x_0)$ is halved. Putting our result for the numerator into our expression for the Bias and noting the cancellation of $f(x_0)$, we have at $x_0 = 0$

$$\operatorname{Bias}\left(\hat{m}(0) : X_{1:n}\right) = hm'(0)2\int_0^1 K(u)udu\left(1 + o_{\mathbb{P}}(1)\right).$$

The steeper the true function near the boundary, the larger the bias is.

We can write out a nice[3] expression for the conditional MSE. Summarizing for 0 at the boundary and x_0 interior,

$$\begin{aligned}\operatorname{MSE}(\hat{m}(0) \mid X_{1:n}) = \Bigg[&\frac{4}{nh}\frac{\sigma^2(0)}{f(0)}\int_0^1 K^2(u)du \\ &+h^2 4m'(0)^2\left(\int_0^1 K(u)udu\right)^2\Bigg]\left(1 + o_{\mathbb{P}}(1)\right)\end{aligned}$$

.

Thus

$$\operatorname{MSE}(\hat{m}(x_0) \mid X_{1:n}) = V(x_0) + \left(\frac{NB}{DB}\right)^2.$$

where

$$V(x_0) = \frac{1}{nh}\frac{\sigma^2(x_0)}{f(x_0)}\int_{-1}^1 K^2(u)du\left(1 + o_{\mathbb{P}}(1)\right).$$

where DB is the KDE of $f(x_0)$ and $DB \xrightarrow{p} f(x_0)$. Using change-of-variables, we found $\mathbb{E}\left[NB\right] = ho(1)$, so $\frac{\mathbb{E}[NB]}{h} \to 0$.

Sharper bias bound using second-order Taylor expansion. When $x_0 = 0$ at the boundary, we only found that $\frac{1}{h}\operatorname{Bias}\left(\hat{m}(0) \mid X_{1:n}\right) \to 0$. Can we get an even sharper characterization of the bias at the boundary? Using now a second-order Taylor

[3]According to some.

expansion,

$$
\begin{aligned}
\mathbb{E}[NB] &= \int_{-1}^{1} K(u)\left(m(x_0+uh)-m(x_0)\right) f\left(x_0+uh\right) du \\
&= \underbrace{\int_{-1}^{1} K(u) m'(x_0) uh f(x_0+uh) du}_{(I)} + \underbrace{\int_{-1}^{1} K(u) \frac{1}{2} m''(\zeta) u^2 h^2 f(x_0+uh) du}_{(II)} \\
(I) &= hm'(x_0) \int_{-1}^{1} K(u) u f(x_0+uh) du \\
&\xrightarrow{h\to 0} hm'(x_0) f(x_0) \int_{-1}^{1} K(u) u du \\
&\xrightarrow{h\to 0} 0 \text{ by assumption} \\
(II) &= \frac{h^2}{2} \int_{-1}^{1} K(u) m''(x_0) u^2 f(x_0) du \\
&\xrightarrow{h\to 0} h^2 f(x_0) m''(x_0) \underbrace{\int_{-1}^{1} u^2 K(u) du}_{\text{known constant by assumption}}.
\end{aligned}
$$

We have shown $(II) = C \cdot h^2 \cdot O_{\mathbb{P}}(1)$, but we have only shown $(I) = ho_{\mathbb{P}}(1)$.
By assumption, f is continuously differentiable at x_0, so we can actually write

$$
\begin{aligned}
(I) &= hm'(x_0)\left[f(x_0) \int_{-1}^{1} uK(u) du + h \int_{-1}^{1} u^2 K(u) f'(\zeta) du \right], \quad \zeta \in (x_0, x_0+uh) \\
&= h^2 m'(x_0) \int_{-1}^{1} u^2 K(u) f'(\zeta) du \\
&= \left[h^2 m'(x_0) f'(x_0) \int_{-1}^{1} u^2 K(u) du \right] (1+o(1)).
\end{aligned}
$$

We can check that $\mathrm{Var}(NB)$ is negligible compared to $\mathbb{E}[NB]$, so we can write

$$
\begin{gathered}
NB = h^2 A(x_0)\left(1+o_{\mathbb{P}}(1)\right) \\
\mathrm{Bias}\left(\hat{m}(x_0) \mid X_{1:n}\right) = \left[h^2 \frac{A(x_0)}{f(x_0)} \right] \left(1+o_{\mathbb{P}}(1)\right).
\end{gathered}
$$

where

$$
A(x_0) = \left(m'(x_0) f'(x_0) + f(x_0) \frac{m''(x_0)}{2} \right) \int_{-1}^{1} u^2 K(u) du.
$$

Summary of MSE bounds. We have derived the following conditional MSE bounds for the Nadaraya-Watson estimator:

Interior ($x_0 \in (0,1)$):

$$\begin{aligned}\mathrm{MSE}(\hat{m}(x_0) \mid X_{1:n}) &= \frac{1}{nh}\frac{\sigma^2(x_0)}{f(x_0)}\int_{-1}^{1} K^2(u)du \\ &+ h^4\left(m'(x_0)\frac{f'(x_0)}{f(x_0)} + \frac{m''(x_0)}{2}\right)^2\left(\int_{-1}^{1} u^2K(u)du\right)^2 \\ &+ o_{\mathbb{P}}\left(\frac{1}{nh} + h^4\right).\end{aligned}$$

Optimal bandwidth: $h^* = n^{-1/5}$ giving MSE rate $n^{-4/5}$.

Boundary ($x_0 = 0$):

$$\begin{aligned}\mathrm{MSE}(\hat{m}(0) \mid X_{1:n}) &= \frac{4}{nh}\frac{\sigma^2(0)}{f(0)}\int_{0}^{1} K^2(u)du \\ &+ h^2 \cdot 4m'(0)^2\left(\int_0^1 K(u)udu\right)^2 \\ &+ o_{\mathbb{P}}\left(\frac{1}{nh} + h^2\right).\end{aligned}$$

Optimal bandwidth: $h^* = n^{-1/3}$ giving MSE rate $n^{-2/3}$.

We see that there is faster convergence in the interior over the boundary because the rate of convergence is of order h^4 versus h^2. Optimally picking $h^* = n^{-\frac{1}{5}}$ in the interior and $h^* = n^{-\frac{1}{3}}$ at the boundary, the MSE is of order $n^{-\frac{4}{5}}$ in the interior and only $n^{-\frac{2}{3}}$ at the boundary.

Nadaraya-Watson adapts to higher order smoothness in the interior, because the smoother m and f are, the sharper characterizations we can get.

In the expression for the MSE in the interior, we wonder why $m'(x_0)\frac{f'(x_0)}{f(x_0)}$ shows up: if the density is changing rapidly at x_0, the MSE is large. Rather than the density being what's important, $\frac{f'(x_0)}{f(x_0)} = (\ln(f(x_0)))'$ is the score function; the rate of change of the log of f at x_0 is what's important. This term is unnecessary and an artifact of using Nadaraya-Watson (called *design bias*) — it's the tradeoff we make for accounting for the smoothness of m and f.

3.4 Comparing NW to Local Linear Regression

Can we achieve a rate better than h^2 at the boundary? We could do this with a kernel that satisfies $\int_0^1 K(u)udu = 0$, which requires a kernel that is negative: $K(u) < 0$ for some u.[4]

However, this would be strange in practice: having different kernels for the boundary and interior, since $h > 0$. People studied "kernel carpentry," which introduces intermediate kernels for x_0 close to the boundary. However, this is not necessary with local linear regression.

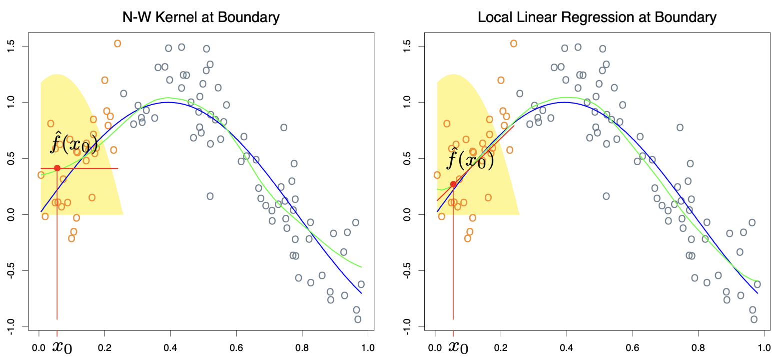
Remark 3.4.1. In local linear regression, we'll start with a standard kernel, pick a bandwidth, and minimize $\frac{1}{n}\sum_i K\left(\frac{X_i - x_0}{h}\right)\left(Y_i - \alpha - \beta\left(X_i - x_0\right)\right)^2$ over α, β. Then

[4] This new condition would be in addition to the assumption: $\int_{-1}^1 K(u)du = 1$ (for consistency of the KDE). The assumption $\int_{-1}^1 K(u)udu = 0$ was only used in analysis of the interior. We can replace this if we are optimizing for the boundary instead.

our estimate of $\mathbb{E}\left[y \mid x = x_0\right] = \hat{m}(x_0) = \hat{\alpha} + \beta(x_0 - x_0) = \hat{\alpha}$.

For example, if K is the rectangular kernel, we do OLS on all observations with $|X_i - x_0| \leq h$.

Figure 3.1: Local linear regression has less boundary bias compared to NW



Supposing $m(X_i) = \alpha + \beta x_i$, the local linear regression has 0 bias, while NW is biased for $\beta \neq 0$. This is because:

- LLR tracks the first order Taylor expansion, accounting for smoothness in that m is approximately linear near x_0.
- NW models $m(x) \approx m(x_0)$.
- In fact, putting $\beta = 0$ into LLR recovers NW, in that

So why use NW if LLR is over a larger class of functions?

$$\min_{\alpha} \frac{1}{n} \sum_{i=1}^{n} K\left(\frac{X_i - x_0}{h}\right)(Y_i - \alpha)$$

  gives $\hat{m}(x_0) = \hat{\alpha}$.

If we redid our previous analysis, replacing NW by local linear regression, we would get in the interior

$$\begin{aligned} \mathrm{MSE}\left(\hat{m}(x_0) \mid X_{1:n}\right) = &\frac{1}{nh}\frac{\sigma^2(x_0)}{f(x_0)}\int_{-1}^{1} K^2(u)du \\ &+ h^4\left(\frac{m''(x_0)}{2}\right)^2\left(\int_{-1}^{1} u^2K(u)du\right)^2(1 + o_{\mathbb{P}}(1)) \end{aligned},$$

so LLR gets the same rate as NW in the interior but without the design bias term of $m'(x_0)\frac{f'(x_0)}{f(x_0)}$. At the boundary, we would have

$$\mathrm{MSE}\left(\hat{m}(0) \mid X_{1:n}\right) = \left(\frac{1}{nh}\frac{\sigma^2(0)}{f(0)}C_1(K) + h^4 m''(0)^2 C_2(K)\right)(1 + o_{\mathbb{P}}(1)),$$

so there is no additional bias at the boundary compared to the interior. We get fast MSE decay without needing kernel carpentry.[5]

[5]Also called automatic kernel carpentry.

Remark 3.4.2 (Interpolation). We say that $\hat{m}$ interpolates the data if $\hat{m}(X_i) = Y_i$ for all i.

The traditional view is that if $\hat{m}$ interpolates the data, it overfits and is a bad estimator. Does this contradict NW? No, because we can have K with a large bump at 0.

Linear Smoothers 4

4.1 Introduction

Recall LLR solves

$$(\hat{\alpha}, \hat{\beta}) = \operatorname*{arg\,min}_{(\alpha,\beta)\in\mathbb{R}^2} \sum_{i=1}^{n} w_i \left(Y_i - \alpha - \beta(X_i - x_0)\right)^2, \qquad w_i \coloneqq K\left(\frac{X_i - x_0}{h}\right).$$

Define

$$\begin{aligned} W &\coloneqq \operatorname{diag}(w_1, \dots, w_n), \\ \tilde{X} &\coloneqq \begin{pmatrix} 1 & X_1 - x_0 \\ \vdots & \vdots \\ 1 & X_n - x_0 \end{pmatrix}, \\ Y &\coloneqq \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \qquad \theta \coloneqq \begin{pmatrix} \alpha \\ \beta \end{pmatrix}. \end{aligned}$$

Then the vector of residuals is

$$r(\theta) \coloneqq Y - \tilde{X}\theta,$$

and the minimand can be written as the quadratic form

$$Q(\theta) \coloneqq \sum_{i=1}^{n} w_i \left(Y_i - \alpha - \beta(X_i - x_0)\right)^2 = r(\theta)' W r(\theta) = (Y - \tilde{X}\theta)' W (Y - \tilde{X}\theta).$$

(If additionally $w_i \geq 0$ for all i so that $W \succeq 0$, then $Q(\theta) = \left\| W^{1/2}(Y - \tilde{X}\theta) \right\|_2^2$.)

Finding the minimizer:

$$\begin{aligned} Q(\theta) &= (Y - \tilde{X}\theta)' W (Y - \tilde{X}\theta) \\ &= Y'WY - 2\theta'\tilde{X}'WY + \theta'\tilde{X}'W\tilde{X}\,\theta, \end{aligned}$$

so

$$\nabla_\theta Q(\theta) = -2\tilde{X}'WY + 2\tilde{X}'W\tilde{X}\,\theta.$$

Setting the gradient to zero gives the normal equations

$$\tilde{X}'W\tilde{X}\,\hat{\theta} = \tilde{X}'WY.$$

When $\tilde{X}'W\tilde{X}$ is invertible,

$$\hat{\theta} = \begin{pmatrix} \hat{\alpha} \\ \hat{\beta} \end{pmatrix} = (\tilde{X}'W\tilde{X})^{-1}\tilde{X}'WY.$$

Define $\hat{m}(x_0) \coloneqq \hat{\alpha}$. Let $e_1 \coloneqq (1,0)'$. Then

$$\hat{m}(x_0) = e_1'\hat{\theta} = e_1'(\tilde{X}'W\tilde{X})^{-1}\tilde{X}'WY = \gamma' Y,$$

where the weight vector (a function of $X_{1:n}, x_0, h$) is

$$\gamma' \coloneqq e_1'(\tilde{X}'W\tilde{X})^{-1}\tilde{X}'W, \qquad \text{so that} \qquad \hat{m}(x_0) = \sum_{i=1}^n \gamma_i Y_i.$$

Equivalently, writing $S_k \coloneqq \sum\limits_{j=1}^n w_j (X_j - x_0)^k$ for $k = 0, 1, 2$, the denominator

$$D \coloneqq S_0 S_2 - S_1^2$$

satisfies $\tilde{X}'W\tilde{X}$ invertible $\iff D \neq 0$, and in that case

$$\gamma_i = \frac{w_i\,(S_2 - (X_i - x_0)S_1)}{D}.$$

We call an estimator of this form (in which the weights do not depend on the outcome variable) a *linear smoother*. Nadaraya-Watson is another linear smoother; we have $\gamma_i = \frac{K\left(\frac{X_i - x_0}{h}\right)}{\sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right)}$. Another linear smoother is k-NN, where $\gamma_i = \frac{1}{k}\mathbb{1}_{X_i \text{ is a neighbor of } x_0}$. The regressogram is also a linear smoother with weights $\gamma_i = \frac{1}{N_j}\mathbb{1}_{X_i \in I_j}$.

One example of an estimator that is not a linear smoother is a regressogram whose bins depend on the values of the Y_is.

Proposition 4.1.1.

We have the following properties of the weights in LLR:

(i) $\sum_i \gamma_i = 1$

(ii) $\sum_{i=1}^n \gamma_i\,(X_i - x_0) = 0$

Proof of Proposition 4.1.1. These could be proved from the direct computation in the formula that characterizes γ_i. However, we will do the proofs in a more clever way, using the definition of a linear smoother.

With

$$\left(\hat{\alpha}, \hat{\beta}\right) = \operatorname*{arg\,min}_{(\alpha,\beta)} \sum_{i=1}^n K\left(\frac{X_i - x_0}{h}\right)\left(\tilde{Y}_i - \alpha - \beta(X_i - x_0)\right)^2,$$

we have that for any $\tilde{Y}_i$, $\hat{\alpha}(\tilde{Y}_1, \ldots, \tilde{Y}_n) = \sum_{i=1}^n \gamma_i \tilde{Y}_i$. We want to pick $\tilde{Y}_i$ in a convenient way and are allowed to do this because of the definition of γ_i does not depend on Y_i; if we can show these properties hold for some vector of outcomes Y and arbitrary regressors, then it must hold for all possible outcomes.

For (i), we use $\tilde{Y}_i = 1$, so the minimization leads to $(\hat{\alpha}, \hat{\beta}) = (1, 0)$. Thus

$$1 = \hat{\alpha} = \sum_{i=1}^n \gamma_i \cdot 1.$$

For (ii), we use $\tilde{Y}_i = X_i - x_0$ to yield

$$\begin{aligned} & \sum_{i=1}^n \gamma_i (X_i - x_0) = \hat{\alpha}\left(X_1 - x_0, \ldots, X_n - x_0\right) \\ & \overset{\text{minimization}}{\Longrightarrow} (\hat{\alpha}, \hat{\beta}) = (0, 1) \\ \Longrightarrow & \sum_{i=1}^n \gamma_i (X_i - x_0) = 0. \end{aligned}$$

Corollary 4.1.2.

By property (ii), we get invariance to subtracting a term linear in deviations from x_0.

Proof of Corollary 4.1.2. Formally, define $\tilde{Y}_i \coloneqq Y_i - m'(x_0)(X_i - x_0)$, and let $m(x) \coloneqq \mathbb{E}\left[Y_i \mid X_i = x\right]$ and $\tilde{m}(x) \coloneqq \mathbb{E}\left[\tilde{Y}_i \mid X_i = x\right]$. Then

$$\begin{aligned} \tilde{m}(x) &= m(x) - m'(x_0)(x - x_0) \\ \tilde{m}'(x_0) &= m'(x_0) - m'(x_0) = 0. \end{aligned}$$

If we fit local linear regression to Y_i to get $\hat{m}(x_0)$ and to $\tilde{Y}_i$ to get $\hat{\tilde{m}}(x_0)$, then

$$\begin{aligned} \hat{\tilde{m}}(x_0) &= \sum_{i=1}^n \gamma_i \tilde{Y}_i \\ &= \sum_{i=1}^n \gamma_i Y_i - m'(x_0) \sum_{i=1}^n \gamma_i (X_i - x_0) \\ &= \sum_{i=1}^n \gamma_i Y_i = \hat{m}(x_0), \end{aligned}$$

where the second-to-last equality follows from property (ii).

This makes sense, because this transformation of Y_i to $\tilde{Y}_i$ should affect the estimation of β, not α.

Remark 4.1.3. The weights can be negative (this is the analog of kernel carpentry in Nadaraya-Watson).

Suppose we are in a one-sided design situation: for all i with $w_i \neq 0$ we have $X_i - x_0 \geq 0$, and at least one such i has $X_i - x_0 > 0$. If all $\gamma_i \geq 0$, then using (i),

$$\sum_{i=1}^{n} \gamma_i (X_i - x_0) \geq 0,$$

and it is strictly > 0 because at least one $(X_i - x_0)$ is positive and has positive weight in the average. This contradicts (ii). Therefore at least one γ_i must be negative.

Recall that with Nadaraya-Watson, we said that we could change the kernel near and at the boundary. Given a linear smoother $\hat{m}(x_0) = \sum_{i=1}^{n} \gamma_i Y_i$ with $\sum_{i=1}^{n} \gamma_i = 1$, we can always write it in Nadaraya-Watson form by defining data-dependent weights $\tilde{K}_i \coloneqq \gamma_i$:

$$\hat{m}(x_0) = \frac{\sum_{i=1}^{n} \tilde{K}_i Y_i}{\sum_{i=1}^{n} \tilde{K}_i}.$$

For LLR, one convenient choice is $\tilde{K}_i = w_i \left(S_2 - (X_i - x_0) S_1\right)$.

4.2 Variations: Local Polynomial Regression

Now consider the lth order polynomial

$$(\hat{\alpha}_0, \ldots, \hat{\alpha}_l)' = \underset{(\alpha_0, \ldots, \alpha_l) \in \mathbb{R}^{l+1}}{\arg\min} \sum_{i=1}^{n} K\left(\frac{X_i - x_0}{h}\right) \left(Y_i - \sum_{j=0}^{l} \alpha_j \frac{(X_i - x_0)^j}{j!}\right)^2,$$

where $l \in \mathbb{N}_0$ and $\hat{m}(x_0) = \hat{\alpha}_0$.

Example 4.2.1.

If $l = 0$, we have NW. If $l = 1$, then we have local linear regression.

The intuition for why we would want to do local polynomial regression is because we may want to fit Taylor polynomials of higher order. With optimal tuning, if we fit an lth degree local polynomial regression and the $\beta \coloneqq l + 1$ derivative of $m(\cdot)$ is bounded and continuous, then the conditional MSE is bounded by a familiar expression:

$$\text{MSE}\left(\hat{m}(x_0) \mid X_{1:n}\right) \lesssim n^{-\frac{2\beta}{2\beta+1}}.$$

As $\beta \to \infty$, the rate approaches $\frac{1}{n}$ from below. Hence, increasing l and our assumption on smoothness gives us faster rates.

So far, we have just looked at the first coefficient $\hat{\alpha}_0$, which represents the point estimate. With a local polynomial regression, we can estimate derivatives. If we want to estimate the νth derivative, we can set $\hat{m}^{(\nu)}(x_0) = \hat{\alpha}_\nu$, if $\nu \leq l$.

Remark 4.2.2. The minimax rate for estimating the derivative is $n^{-\frac{2(\beta-\nu)}{2\beta+1}}$, which is slower than the rate for the point estimate. This implies that estimating derivatives is harder the higher the order of the derivative is.

As a rule of thumb, to estimate the νth derivative,[1] we can use a local polynomial of degree $l = \nu + 1$. For example, for a point estimate, we would want to use LLR (and not NW), and if we wanted to estimate a derivative, we would use a quadratic. This rule of thumb was actually rigorously considered.

4.3 Local Polynomial Regression for Density Estimation

Recall that we have done density estimation using the histogram estimator and KDE. KDE gave boundary bias when we did not change the kernel depending on x_0. But, if we allow $K(u) < 0$ for some u,[2] then it could be the case that

$$\hat{f}(x_0) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{X_i - x_0}{h}\right) < 0.$$

The solution to this problem is to clip $\hat{f}(x_0)$ to be in $[0, \infty)$ and rescaling so it integrates to 1.

Can we use an analog of local polynomial regression for density estimation? The conceptual difficulty is that in a density estimation problem, we don't have Y_i.

Classical Idea:

(i) Use KDE or a histogram estimator with a very small h and put $Y_i = \hat{f}(X_i)$ (an estimate of density that is heavily undersmoothed.)

(ii) Then do local polynomial regression with "optimal" h.

This works because undersmoothing makes the first-stage bias negligible: for fixed x, $\mathbb{E}\left[\hat{f}(x)\right] \approx f(x)$, and taking $\tilde{h}$ small keeps that bias small compared to the second-stage error.

However, this approach presents a difficulty of needing to choose two bandwidths.

We have a better approach through the ECDF: while we don't know any estimator of the density that does not require a bandwidth, we have seen the ECDF as an estimator of the distribution function that does not rely on a bandwidth. We can use local polynomial regression on (Y_i, X_i) given by

$$Y_i = \hat{F}(X_i) = \frac{1}{n} \sum_{j=1}^{n} \mathbb{1}_{X_j \leq X_i}$$

to find the derivative (which would be the density!).

4.4 Choosing Hyperparameters

So far, we have been choosing h to equalize the orders of n of the variance and square bias in the MSE.

In the interior, we have the MSE of the Nadaraya-Watson estimator is bounded

[1] With the goal of getting an accurate estimate without making too many assumptions on the derivatives of $m(\cdot)$.

[2] For $\int_0^1 K(u)du = 0$ to hold so we can get rid of the extra bias at the boundary, we need $K(u) < 0$ somewhere.

with

$$\begin{aligned}\mathrm{MSE}(\hat{m}(x_0) \mid X_{1:n}) &= \frac{1}{nh}\frac{\sigma^2(x_0)}{f(x_0)}\int_{-1}^{1} K^2(u)du \\ &+ h^4\left(m'(x_0)\frac{f'(x_0)}{f(x_0)} + \frac{m''(x_0)}{2}\right)^2\left(\int_{-1}^{1} u^2K(u)du\right)^2 \\ &+ o_{\mathbb{P}}\left(\frac{1}{nh} + h^4\right).\end{aligned}$$

Taking a derivative with respect to h to retain the constants, this yields an optimal bandwidth of

$$h^* = \left(\frac{1}{n}\frac{\frac{\sigma^2(x_0)}{f(x_0)}\int_{-1}^{1} K^2(u)du}{\left(m'(x_0)\frac{f'(x_0)}{f(x_0)} + \frac{m''(x_0)}{2}\right)^2\left(\int_{-1}^{1} u^2K(u)du\right)^2}\right)^{\frac{1}{5}}.$$

For theory, we would say $h^* \asymp n^{-\frac{1}{5}}$, but in practice, the constant is (more) important.

Plugging in h^* yields $MSE \asymp n^{-\frac{4}{5}}$. But how should we choose the kernel? The MSE depends on the quantity $(*)\frac{\int_{-1}^{1} K^2(u)du}{\left(\int_{-1}^{1} u^2K(u)du\right)^2}$; we can minimize again over all kernels that satisfy some requirements, which may include:

- symmetry
- nonnegativity
- continuity
- $\int_{-1}^{1} K(u)du = 1$.

With this expansion/bound on the MSE, we can show the Epanechnikov kernel is optimal. However, in practice, the choice of kernel is not that important.

Does this mean that (*) is relatively constant across different standard kernels?

Looking at our expression for h^*, we don't actually know $\sigma^2(x_0)$, $f(x_0)$, nor $m''(x_0)$. So, we should replace these with estimated versions $\hat{\sigma}^2(x_0), \hat{f}(x_0), \hat{m}''(x_0)$. However, to get $\hat{h}$ to estimate $m(x_0)$, we need an estimate of $m''(x_0)$, which is *more* difficult to estimate.

The argument in favor of this says that it's not catastrophic if we estimate $m''(x_0)$ poorly. We would pick some initial bandwidth to estimate $\hat{m}''(x_0)$ by a heuristic, then get $\hat{h}^*$, so we can get $\hat{m}(x_0)$. This is called the "plug-in" bandwidth selection. Despite the several estimation steps, this still turns out to usually be better than saying the constant in the expression for h^* is 1.

Minimax Plug-In: Suppose that we have estimates $\hat{f}(x_0)$ and $\hat{\sigma}^2(x_0)$. The minimax approach is to bound the worst-case MSE over $\mathbb{P}$ when $m \in \mathscr{G}(2, C)$, where

$$\mathscr{G}(2, C) \coloneqq \left\{m : \sup_{x\in[0,1]} |m''(x)| \le C\right\}.$$

Heuristically,[3] we should have the bound

$$\sup_{m\in\mathscr{G}(2,C)} \operatorname{MSE}\left(\hat{m}(x_0)\mid X_{1:n}\right) \le \frac{1}{nh}\frac{\sigma^2(x_0)}{f(x_0)}\int_{-1}^{1} K^2(u)du + h^4\left(m'(x_0)\frac{f'(x_0)}{f(x_0)}+\frac{C}{2}\right)^2\left(\int_{-1}^{1}u^2K(u)du\right)^2,$$

so the optimal h^* is

$$h^* = \left(\frac{1}{n}\frac{\frac{\sigma^2(x_0)}{f(x_0)}\int_{-1}^{1}K^2(u)du}{\left(m'(x_0)\frac{f'(x_0)}{f(x_0)}+\frac{C}{2}\right)^2\left(\int_{-1}^{1}u^2K(u)du\right)^2}\right)^{\frac{1}{5}}.$$

To make this a genuine bound, one typically also controls the $m'(x_0)\frac{f'(x_0)}{f(x_0)}$ term, e.g. by assuming a uniform design or an additional bound on m'.

and it might be better to overestimate $m''(x_0)$ (when our metric is the maximum risk).

But how would we actually pick C?

(i) In the genuine minimax approach, some domain analysis would let you pick C.

(ii) Sensitivity analysis might specify plausible upper bounds $C_1,\dots,C_n$, and conduct analysis separately for each C_i. Then we could see how the assumptions affect our bandwidth (and thus, once we get to inference, the width of our confidence intervals).

(iii) (Armstrong-Kolesat; Heuristics) We could globally fit a polynomial $\hat{p}$ of some degree (say, $l=4$) and put $C=\sup_{x\in[0,1]}|p''(x_0)|$.

Thus, creating an upper bound on $m''(x_0)$ might be better than actually trying to estimate $m''(x_0)$. Improvements of this method include:

- We won't need an estimate of $f(x_0)$
- Works for any linear smoother
- Works for any convex class $\mathscr{M}$.

Consider a linear smoother $\hat{m}(x_0)=\sum_i \gamma_i Y_i$, where $\gamma_i=\gamma_i(h)$. Assume the following:

(i) $\operatorname{Var}(Y_i\mid X_i)\le\sigma^2$

(ii) $m(\cdot)\in\mathscr{M}$, where $\mathscr{M}$ is some class like $\mathscr{G}(2,C)$.

We can write

$$\begin{aligned}
\operatorname{MSE}\left(\hat{m}(x_0)\mid X_{1:n}\right) &= \operatorname{Var}\left(\hat{m}(x_0)\mid X_{1:n}\right)+\operatorname{Bias}^2\left(\hat{m}(x_0)\mid X_{1:n}\right)\\
\operatorname{Var}\left(\hat{m}(x_0)\mid X_{1:n}\right) &= \sum_{i=1}^n \gamma_i^2\operatorname{Var}\left(Y_i\mid X_i\right)\le\sigma^2\sum_{i=1}^n\gamma_i^2\\
\operatorname{Bias}\left(\hat{m}(x_0)\mid X_{1:n}\right) &= \mathbb{E}\left[\hat{m}(x_0)\mid X_{1:n}\right]-m(x_0)\\
\text{since } \gamma_i \text{ is a function of } X_{1:n},\quad &= \sum_{i=1}^n\gamma_i m(X_i)-m(x_0)\\
\sup_{m\in\mathscr{M}}\operatorname{MSE}\left(\hat{m}(x_0)\mid X_{1:n}\right) &\le \sigma^2\sum_{i=1}^n\gamma_i^2+\sup_{m\in\mathscr{M}}\left(\sum_{i=1}^n\gamma_i m(X_i)-m(x_0)\right)^2,
\end{aligned}$$

[3] Why is the proof not immediate?

with equality if we have homoskedasticity ($\mathrm{Var}(Y_i \mid X_i) = \sigma^2, \forall\, i$.

Note that we have no more dependency on m because of our assumption $m \in \mathscr{M}$ and the sup. Furthermore, the mapping $h \mapsto \gamma_i(h)$ is known by running LLR with bandwidth h. Hence, with $\gamma_i = \gamma_i(h)$, we can minimize the RHS over h to get $\hat{h}$ as an estimate of h^*.

The difficult part of minimizing the RHS is computing the sup over all $m \in \mathscr{M}$. Sometimes we can do this analytically (for instance, the SY class), but this is rare. Instead, we can use tools from convex optimization to solve/compute $\sup_{m\in\mathscr{M}} \left(\sum_{i=1}^{n} \gamma_i(h) m(X_i) - m(x_0) \right)$.

This task is not easy, so we explore it in the next section.

4.5 Computing the Minimax-Optimal Bandwidth

Supposing for now that we want to pick h^* from the set $\{h_1, \dots, h_N\}$. Considering

$$\phi(h) = \sigma^2 \sum_{i=1}^{n} \gamma_i(h)^2 + \sup_{m\in\mathscr{M}} \left| \sum_{i=1}^{n} \gamma_i(h) m(X_i) - m(x_0) \right|^2,$$

how can we compute

$$\sup_{m\in\mathscr{M}} \left| \sum_{i=1}^{n} \gamma_i(h) m(X_i) - m(x_0) \right| ?$$

(i) The classical approach is to do this analytically — which is not impossible with classes like $SY_{x_0}(\beta, L)$.

(ii) We can also use optimization, which is particularly convenient when $\mathscr{M}$ is convex.

Definition 4.5.1 (Convex set of functions).

$\mathscr{M}$ is convex if for any $m_1, m_2 \in \mathscr{M}$ and $\lambda \in [0,1]$,

$$\lambda m_1 + (1-\lambda) m_2 \in \mathscr{M}.$$

Examples include the class of all polynomials up to degree K (too small for our use), continuous functions (too large), and L-Lipschitz functions.

Splitting up our problem, we can compute

$$\max \left\{ \sup_{m\in\mathscr{M}} \sum_{i=1}^{n} \gamma_i(h) m(X_i) - m(x_0),\ \sup_{m\in\mathscr{M}} m(x_0) - \sum_{i=1}^{n} \gamma_i(h) m(X_i) \right\}. \tag{4.1}$$

This is actually a finite-dimensional optimization problem. Defining

$$\tilde{\mathscr{M}} = \left\{ \begin{pmatrix} m(X_1) \\ \vdots \\ m(X_n) \\ m(x_0) \end{pmatrix} : m \in \mathscr{M} \right\} \subseteq \mathbb{R}^{n+1},$$

we can check that $\mathscr{M}$ being convex implies $\tilde{\mathscr{M}}$ is convex, so it suffices to solve

$$\max \sum_{i=1}^{n} \gamma_i(h) m(X_i) - m(x_0)$$

such that

$$\tilde{m} = \begin{pmatrix} m(X_1) \\ \vdots \\ m(X_n) \\ m(x_0) \end{pmatrix} \in \tilde{\mathscr{M}}.$$

The maximand is linear in $\tilde{m}$ and the constraint is a convex constraint on $\tilde{m}$.

A convex optimization problem is one in which we are minimizing a (weakly?) convex function subject to a constraint defined by a convex set. Software like CVXR can help us determine if these problems are solvable and solve them.

With the more modern approach that uses computation, we can get exact values instead of the usual upper bound provided by analytical methods.

To minimize $\psi(h)$ over $\{h_1, \ldots, h_J\}$,[4] we need to solve $2J$ optimization problems. If the two terms in the problems (4.1) are equal, we just need to solve J optimization problems. In particular, this happens when $\mathscr{M}$ is closed under negation.

By taking an outer min over some classes, we don't have to limit ourselves to linear smoothers of one type. For instance, we could optimally tune k-NN and LLR, then optimize over these estimator classes. But why not take the outer min over $\gamma \in \mathbb{R}^n$, instead of just the weights corresponding to the particular estimators we know? Computing

$$\min_{\gamma \in \mathbb{R}^n} \sup_{m \in \mathscr{M}} \psi(\gamma, m)$$

is a formidable task, since we have nested optimization in this min-max problem.

If we can't solve the inner sup analytically, we can sometimes use Lagrangian duality in a single convex optimization problem, or a more powerful framework inspired from Donoho (1994) that reduces the problem to a sequence of convex optimization problems.

While outside the scope of this class, we recognize that this computational approach exists.

Implications for Choice of Linear Smoothers.

If we are minimizing the objective over all $\gamma \in \mathbb{R}^n$, then do the weights satisfy properties of LLR and NW we saw earlier (like summing to 1)?

Example 4.5.2.

If $\sum_i \gamma_i \neq 1$, then considering $\mathscr{M} = \Sigma(1, L)$, then the risk $\psi(\gamma, m)$ is unbounded by considering $m(\cdot) = c$ and sending $c \to \infty$. Thus, the optimization forces $\sum_i \gamma_i = 1$.
Similarly, if $\mathscr{M} = \Sigma(2, L)$, then we get

$$\sum_i \gamma_i (X_i - x_0) = 0,$$

[4] Note the optimal bandwidth is chosen by computing the MSE, not just the worst-case bias.

as the case was with LLR.

Implications for Inference. Recall that in nonparametrics, bias is nonnegligible compared to standard errors. We saw that undersmoothing (increasing variance, lowering bias), bias correction, and ignoring bias were ways of addressing this. We now discuss bias-aware inference.

To focus only on bias, we assume that $Y_i \mid X_i \sim N(m(X_i), \sigma_i^2)$ for known σ_i^2 and consider

$$\hat{m}(x_0) = \sum_{i=1}^{n} \gamma_i Y_i.$$

Then

$$\hat{m}(x_0) \mid X_{1:n} \sim N\left(\sum_{i=1}^{n} \gamma_i m(X_i), \sum_{i=1}^{n} \gamma_i^2 \sigma_i^2\right),$$

so

$$\hat{m}(x_0) - m(x_0) \mid X_{1:n} \sim N\left(\operatorname{Bias}\left(\hat{m}(x_0) \mid X_{1:n}\right), \operatorname{Var}\left(\hat{m}(x_0) \mid X_{1:n}\right)\right).$$

In the usual case where $\frac{\operatorname{Bias}(\hat{m}(x_0) \mid X_{1:n})}{\operatorname{se}(\hat{m}(x_0) \mid X_{1:n})} \approx 0$,

$$\hat{m}(x_0) - m(x_0) \mid X_{1:n} \sim N\left(0, \operatorname{Var}\left(\hat{m}(x_0) \mid X_{1:n}\right)\right)$$

yields the confidence interval

$$I = \left[\hat{m}(x_0) \pm z_{1-\frac{\alpha}{2}} \operatorname{se}\left(\hat{m}(x_0) \mid X_{1:n}\right)\right]$$

and probabilistic statement

$$\mathbb{P}\left[m(x_0) \in I \mid X_{1:n}\right] = 1 - \alpha.$$

But since we don't have

$$\frac{\operatorname{Bias}\left(\hat{m}(x_0) \mid X_{1:n}\right)}{\operatorname{se}\left(\hat{m}(x_0) \mid X_{1:n}\right)} \approx 0,$$

we instead use a bias bound. For $m \in \mathscr{M}$,

$$\begin{aligned} \left|\operatorname{Bias}\left(\hat{m}(x_0) \mid X_{1:n}\right)\right| &= \left|\sum_{i=1}^{n} \gamma_i m(X_i) - m(x_0)\right| \\ &\leq B(x_0) \coloneqq \sup_{m \in \mathscr{M}} \left|\sum_{i=1}^{n} \gamma_i m(X_i) - m(x_0)\right|. \end{aligned}$$

Hence, a conservative two-sided interval is

$$I_{\text{naive}} = \left[\hat{m}(x_0) \pm \left(z_{1-\frac{\alpha}{2}} \operatorname{se}\left(\hat{m}(x_0) \mid X_{1:n}\right) + B(x_0)\right)\right],$$

which satisfies $\mathbb{P}\left[m(x_0) \in I_{\text{naive}} \mid X_{1:n}\right] \geq 1 - \alpha$. In a one-sided test, the analogous upper interval is

$$I_{\text{naive}}^{+} = \left(-\infty, \, \hat{m}(x_0) + z_{1-\alpha} \operatorname{se}\left(\hat{m}(x_0) \mid X_{1:n}\right) + B(x_0)\right].$$

This approach can be conservative (often strictly) when $B(x_0) > 0$.

For two-sided inference, one can replace the additive $+B(x_0)$ inflation by a bias-aware critical value. Let $Z \sim N(0,1)$,

$$\tilde{b} := \frac{B(x_0)}{\operatorname{se}\left(\hat{m}(x_0) \mid X_{1:n}\right)} \quad \text{and} \quad \chi(\tilde{b}) := \inf\left\{t : \inf_{|b|\le\tilde{b}} \mathbb{P}\left[|Z+b| \le t\right] \ge 1-\alpha\right\}.$$

Then

$$I = \left[\hat{m}(x_0) \pm \chi(\tilde{b}) \operatorname{se}\left(\hat{m}(x_0) \mid X_{1:n}\right)\right]$$

satisfies $\inf_{m\in\mathscr{M}} \mathbb{P}\left[m(x_0) \in I \mid X_{1:n}\right] \ge 1-\alpha$ and has a shorter half-length than the naive interval:

$$\chi(\tilde{b}) \operatorname{se}\left(\hat{m}(x_0) \mid X_{1:n}\right) \le z_{1-\frac{\alpha}{2}} \operatorname{se}\left(\hat{m}(x_0) \mid X_{1:n}\right) + B(x_0).$$

Moreover, $\chi(0) = z_{1-\frac{\alpha}{2}}$ and $\chi(\tilde{b}) \approx \tilde{b} + z_{1-\alpha}$ for large $\tilde{b}$. Relaxing the normality assumption. Recall we assumed that $Y_i \mid X_i \sim N(m(X_i), \sigma_i^2$ What happens if we relax this normality assumption? Suppose $(Z_i)_{i=1}^n$ is a sequence of *i.i.d.* variables and put $U = \sum_{i=1}^n Z_i$. Define

$$\mu_i := \mathbb{E}\left[Z_i\right], \quad \tau_i^2 := \operatorname{Var}\left(Z_i\right), \quad \rho_i := \mathbb{E}\left[\left|Z_i - \mu_i\right|^3\right].$$

We know

$$\mathbb{E}\left[U\right] = \sum_{i=1}^n \mu_i, \quad \operatorname{Var}\left(U\right) = \sum_{i=1}^n \tau_i^2, \quad \frac{U - \mathbb{E}\left[U\right]}{\sqrt{\operatorname{Var}\left(U\right)}} \approx N(0,1).$$

Theorem 4.5.3 (Berry-Esseen).

For $\Phi(t)$ the standard normal CDF, we have

$$\sup_{t\in\mathbb{R}} \left|\mathbb{P}\left[\frac{U - \mathbb{E}\left[U\right]}{\sqrt{\operatorname{Var}\left(U\right)}} \le t\right] - \Phi(t)\right| \le C \frac{\sum_{i=1}^n \rho_i}{\left(\operatorname{Var}\left(U\right)\right)^{\frac{3}{2}}}$$

for some universal constant C.
In particular, if the upper bound goes to 0, then the asymptotic distribution is approximately $N(0,1)$.

Example 4.5.4.

Consider the case where Z_i are i.i.d. with $\rho := \mathbb{E}\left[\left|Z_1 - \mathbb{E}\left[Z_1\right]\right|^3\right]$ and $\tau^2 := \operatorname{Var}\left(Z_1\right)$. Then the bound becomes

$$\sup_{t\in\mathbb{R}} \left|\mathbb{P}\left[\frac{U - \mathbb{E}\left[U\right]}{\sqrt{\operatorname{Var}\left(U\right)}} \le t\right] - \Phi(t)\right| \le C \frac{n\rho}{\left(n\tau^2\right)^{3/2}} = C \frac{\rho}{\sqrt{n}\tau^3}.$$

We can apply this to linear smoothers by taking, conditional on $X_{1:n}$,

$$Z_i \coloneqq \gamma_i \left(Y_i - m(X_i)\right), \quad \mu_i = 0, \quad \begin{aligned} \tau_i^2 &= \gamma_i^2 \operatorname{Var}\left(Y_i \mid X_i\right), \\ \rho_i &= |\gamma_i|^3 \, \mathbb{E}\left[|Y_i - m(X_i)|^3 \mid X_i\right]. \end{aligned}$$

along with the following sufficient assumptions:

(i) $\sigma_i^2 = \operatorname{Var}\left(Y_i \mid X_i\right) \geq c > 0$ for all i,

(ii) $\mathbb{E}\left[|Y_i - m(X_i)|^3 \mid X_i\right] \leq C < \infty$ for all i, and

(iii) $\frac{\max_i \gamma_i^2}{\sum_{i=1}^n \gamma_i^2} \xrightarrow{n\to\infty} 0$ almost surely

to get that

$$\sup_{t\in\mathbb{R}} \left| \mathbb{P}\left[\frac{\hat{m}(x_0) - \mathbb{E}\left[\hat{m}(x_0) \mid X_{1:n}\right]}{\sqrt{\operatorname{Var}\left(\hat{m}(x_0) \mid X_{1:n}\right)}} \leq t \mid X_{1:n}\right] - \Phi(t) \right| \xrightarrow{n\to\infty} 0.$$

Before, with conditional normality of $Y_i \mid X_i$, we had the exact conditional distribution

$$\hat{m}(x_0) \mid X_{1:n} \sim N\left(\mathbb{E}\left[\hat{m}(x_0) \mid X_{1:n}\right], \operatorname{Var}\left(\hat{m}(x_0) \mid X_{1:n}\right)\right).$$

Under the Berry-Esseen conditions above, we instead have the asymptotic approximation

$$\frac{\hat{m}(x_0) - \mathbb{E}\left[\hat{m}(x_0) \mid X_{1:n}\right]}{\sqrt{\operatorname{Var}\left(\hat{m}(x_0) \mid X_{1:n}\right)}} \approx N(0,1).$$

Relaxing the variance assumption. We used our assumption of known σ_i^2 in two places:

- Choosing the bandwidth from

$$\sum_{i=1}^n \sigma_i^2 \gamma_i^2 + \sup_{m\in\mathscr{M}} \left| \sum_{i=1}^n \gamma_i m(X_i) - m(x_0) \right|^2.$$

- Computing the CI in

$$\hat{m}(x_0) \pm \tilde{\chi} \operatorname{se}\left(\hat{m}(x_0) \mid X_{1:n}\right),$$

  as we need knowledge of

$$\operatorname{se}\left(\hat{m}(x_0) \mid X_{1:n}\right)^2 = \sum_{i=1}^n \gamma_i^2 \sigma_i^2.$$

Plugging in a rough guess into the first could still give a linear smoother, while a rough guess for σ_i in the second could result in incorrect coverage.

So what should we do if we don't know σ_i^2? To avoid undercoverage, it might be okay to upper bound σ_i^2 and inflate the CI by even more. We can estimate

Consider $\tilde{\sigma}_i^2 = \frac{\sigma_i^2}{2}$. Then square bias is weighted twice as much in the bias-variance tradeoff, but coverage is much less than $1-\alpha$.

$\sigma_i^2 = \operatorname{Var}(Y_i \mid X_i)$ with $\hat{\sigma}_i^2$ as the sample variance to get

$$\widehat{\operatorname{Var}}(\hat{m}(x_0) \mid X_{1:n}) = \sum_{i=1}^{n} \gamma_i^2 \hat{\sigma}_i^2.$$

We want $\hat{\sigma}_i^2$ to satisfy:

(i) Consistency (universal, in particular), so

$$\max_i \left| \frac{\hat{\sigma}_i^2}{\sigma_i^2} - 1 \right| \xrightarrow{\mathbb{P}} 0$$

yields

$$\frac{\widehat{\operatorname{Var}}(\hat{m}(x_0) \mid X_{1:n})}{\operatorname{Var}(\hat{m}(x_0) \mid X_{1:n})} \xrightarrow{\mathbb{P}} 1.$$

For a consistent estimator of σ_i^2, consider

(i) $\hat{\epsilon}_i = Y_i - m(X_i)$, run k-NN on $\hat{\epsilon}_i$, and put $\hat{\sigma}_i^2$ as the estimator of $\hat{\epsilon}_i^2$.

(ii) Write

$$\operatorname{Var}(Y_i \mid X_i) = \mathbb{E}\left[Y_i^2 \mid X_i\right] - m(X_i)^2$$

as the difference of nonparametric regressions of Y_i^2 and Y_i on X_i.

But actually, we don't need consistency. Estimating $\operatorname{Var}(Y_i \mid X_i)$ is harder than estimating $m(X_i)$ (our original goal was to estimate $m(x_0)$, so we would like to avoid estimating variance in the process), so we can estimate

$$\hat{\operatorname{Var}}(\hat{m}(x_0) \mid X_{1:n}) = \sum_{i=1}^{n} \gamma_i^2 \hat{\sigma}_i^2,$$

where $\hat{\sigma}_i^2$ is not necessarily consistent for σ_i^2 but is nearly unbiased. By a LLN argument, we would have that this variance estimator is consistent.

This is analogous to the idea of Eicker-Huber-White, or heteroskedasticity-robust standard errors in linear regression.

Example 4.5.5.

We can implement this with

$$\hat{\sigma}_i^2 = (Y_i - \hat{m}(X_i))^2$$

and $\hat{m}$ consistent.

But we don't even need consistency of $\hat{m}$, following the k-NN estimator of Abadie and Imbens (2005). Fixing J (e.g. $J = 3$), we define

$$N_J(X_i) = \{J \text{ nearest neighbors of } X_i \text{ excluding itself}\}$$

and estimate

$$\hat{\sigma}_i^2 = \left(Y_i - \frac{1}{J} \sum_{j \in N_J(x_i)} Y_j \right)^2.$$

We see by centering on $m(X_i)$, if there are J ties, where $X_j = X_i$ for J js,

$$\begin{aligned}\mathbb{E}\left[\hat{\sigma}_i^2 \mid X_{1:n}\right] &= \mathrm{Var}\left(Y_i \mid X_i\right) + \mathrm{Var}\left(\frac{1}{J}\sum_{j\in N_J(X_i)} Y_j \mid X_{1:n}\right) \\ &= \sigma_i^2\left(1+\frac{1}{J}\right) \\ &= \sigma_i^2 \frac{J+1}{J},\end{aligned}$$

so we should actually estimate

$$\hat{\sigma}_i^2 = \frac{J}{J+1}\left(Y_i - \frac{1}{J}\sum_{j\in N_J(x_i)} Y_j\right)^2.$$

Note that the individual estimates are themselves not consistent, but their lack of bias gives us consistency after averaging with weights γ_i^2.
Finally, we can fit some global linear regression

$$\begin{aligned}Y_i &= \beta_0 + \beta_1 X_i \\ \hat{\epsilon}_i &= Y_i - \hat{\beta}_0 + \hat{\beta}_1 X_i,\end{aligned}$$

with $\hat{\sigma}_i^2 = \hat{\epsilon}_i^2$, so

$$\mathbb{E}\left[\hat{\sigma}_i^2 \mid X_{1:n}\right] - \sigma_i^2 \geq 0$$

for large n and we have *at least* the coverage we intended to have. (This comes at the expense of conservative inference.)

What properties of the confidence interval

$$\left(-\infty, \hat{m}(x_0) + z_{1-\alpha}\sqrt{\widehat{\mathrm{Var}}(\hat{m}(x_0))} + B(x_0)\right)$$

leads to appropriate asymptotic coverage of

$$\lim_{n\to\infty} \mathbb{P}\left[m(x_0) \in I\right] \geq 1-\alpha?$$

We need

$$\frac{\widehat{\mathrm{Var}}(\hat{m}(x_0) \mid X_{1:n})}{\mathrm{Var}\left(m(x_0) \mid X_{1:n}\right)} \geq 1$$

and asymptotic normality of $\hat{m}(x_0) \mid X_{1:n}$.

This may work even if $\hat{m}(x_0)$ is not consistent for $m(x_0)$ and $\mathrm{Var}\left(\hat{m}(x_0) \mid X_{1:n}\right)$ because we are inflating the CI by $B(x_0)$, which for each n, acts as an upper bound on the bias. While this seems like a trivial result, this is important in problems that are partially identified: problems in which even with $n=\infty$, it is impossible to be consistent (since most other approaches to inference would fail).

Consider the deterministic estimator $\tilde{m}(x_0)$. Since $|m(x_0) - \tilde{m}(x_0)| \leq B(x_0)$, we have that $m(x_0) \in [\tilde{m}(x_0) \pm B(x_0)]$.

Example 4.5.6 (Partial identification).

Suppose all $X_i \geq 1$ but we want to estimate $m(0)$. We could make the following assumptions to be consistent in estimation $m\left(0\right)$:

- $m\left(0\right) = m(1)$, and we can estimate $m(1)$.
- Impose a functional form: suppose $m(\cdot)$ is linear or a polynomial.

These are very strong assumptions (at least in nonparametrics).
More reasonably assuming something like 1-Lipschitz continuity, we could never be consistent for $m(0)$ but we can get an informative bound: asymptotically, we would think $m(0) \in [m(1) \pm 1]$,[a] which we could estimate with $\hat{m}(1)$ and hence report $m(0) \in [\hat{m}(1) \pm t]$, where t is determined by a bias-aware approach.

[a] This interval is called a population partial identification interval.

Remark 4.5.7. Recall that we had

$$\lim_{n\to\infty} \mathbb{P}\left[m(x_0) \in I\right] \geq 1-\alpha.$$

In fact, the actual guarantee for bias-aware inference is

$$\liminf_{n\to\infty} \inf_{m\in\mathscr{M}} \mathbb{P}\left[m(x_0) \in I \mid X_{1:n}\right] \geq 1-\alpha,$$

which is called a distribution-uniform bound/coverage.

However, considering

$$\inf_{m\in\mathscr{M}} \liminf_{n\to\infty} \mathbb{P}\left[m(x_0) \in I \mid X_{1:n}\right] \geq 1-\alpha,$$

the first bound is stronger (and implies) the second, as the first one is a *uniform* lower bound across $\mathscr{M}$ on $\mathbb{P}\left[m(x_0) \in I\right]$, while the second is a *pointwise* notion.

The second bound arises from bias-correction methods, which does not force us to pick $\mathscr{M}$.[5]

[5] Bias-correction methods give us larger CIs, I think.

Causal Inference 5

5.1 Rubin Causal Model

Consider covariates $\{X_i\}$ and treatment indicators $\{W_i\}$, along with the response variable $\{Y_i\}$. The guiding question is: Does W_i "causally impact" Y_i?

Example 5.1.1.

Let W_i be an indicator for the event that student i received a certificate of merit and $Y_i \in \{0, 1\}$ be an indicator for if student i received a scholarship within a year of the certifications being given out.
A naive starting point is to conduct a difference-in-means test: considering

$$\hat{\tau} = \frac{1}{\#\{W_i = 1\}} \sum_{i:W_i=1} Y_i - \frac{1}{\#\{W_i = 0\}} \sum_{i:W_i=0} Y_i,$$

if $\hat{\tau}$ is much larger than 0, we may want to conclude that there is a causal effect of W_i on Y_i.
However, because confounders like previous academic performance influence both W_i and Y_i, we cannot immediately say that the relation is causal.

We want to develop a formal language to talk about causality. While there are many possibility, we will use the potential outcomes framework of Neyman and Rubin.

Definition 5.1.2 (Potential outcomes).

The potential outcome if Y_i is treated is $Y_i(1)$ and $Y_i(0)$ if Y_i is not treated.
For each student, we imagine scenarios in which individual i gets treatment and where they don't get treatment. We link potential outcomes to the observed Y_i by writing

$$Y_i = \begin{cases} Y_i(0), & W_i = 0 \\ Y_i(1), & W_i = 1 \end{cases} = W_i Y_i + (1 - W_i) Y_i.$$

Implicitly, we are assuming SUTVA.

SUTVA stipulates:

(i) Treatment of i only depends on treatment assignment of i and not other observations

(ii) i's potential outcomes are affected by its treatment assignment and treatment, and not those of others

The fundamental challenge of causal inference is that we only see only of the two potential outcomes, and not both.

Definition 5.1.3 (Individual treatment effect).

The individual treatment effect is

$$\tau_i = Y_i(1) - Y_i(0).$$

While we do not know τ_i, we can use it to define estimands of interest. If we assume $\mathbb{P}\left[W_i = 1\right] \in (0,1)$, then we see

$$\begin{aligned}
\hat{\tau} &= \frac{1}{\#\left\{i : W_i = 1\right\}} \sum_{i:W_i=1} Y_i - \frac{1}{\#\left\{i : W_i = 0\right\}} \sum_{i:W_i=0} Y_i \\
&\xrightarrow{\mathbb{P}} \mathbb{E}\left[Y_i \mid W_i = 1\right] - \mathbb{E}\left[Y_i \mid W_i = 0\right] \\
&= \mathbb{E}\left[Y_i(1) \mid W_i = 1\right] - \mathbb{E}\left[Y_i(0) \mid W_i = 0\right] \\
&= \underbrace{\mathbb{E}\left[Y_i(1) - Y_i(0) \mid W_i = 1\right]}_{ATT} + \underbrace{\left(\mathbb{E}\left[Y_i(0) \mid W_i = 1\right] - \mathbb{E}\left[Y_i(0) \mid W_i - 0\right]\right)}_{\text{confounding}} \\
&= \underbrace{\mathbb{E}\left[Y_i(1) - Y_i(0) \mid W_i = 0\right]}_{\text{ATU}} + \left(\mathbb{E}\left[Y_i(1) \mid W_i = 1\right] - \mathbb{E}\left[Y_i(1) \mid W_i = 0\right]\right).
\end{aligned}$$

With the very strong assumption of $W_i \perp\!\!\!\perp (Y_i(0), Y_i(1))$, we would have that $\hat{\tau} = \mathbb{E}\left[Y_i(1) - Y_i(0)\right] = ATE$. Sometimes this orthogonality is enforced by construction; e.g. in RCTs and A/B tests. However, we often don't have this because of ethical reasons, monetary constraints, or lack of control over treatment assignment in existing datasets (observational studies).

Example 5.1.4 (Regression Discontinuity Design).

Continuing with the certificate of merit example, consider covariates $X_i \in \{1, \ldots, 20\}$ as test scores that determine W_i by

$$W_i = \mathbb{1}_{X_i > c}, \quad c = 10.$$

We call X_i the *running variable*. The idea of RDD is that students who have $X_i = 10$ and students who have $X_j = 11$ are similar and comparable.

How can we formalize the notion that observations near some cutoff for receiving treatment vs. no treatment are as if they were randomized? One method is the continuity-based approach. Define the conditional response functions

$$\mu_{(1)} = \mathbb{E}\left[Y_i(1) \mid X_i = x\right], \quad \mu_{(0)}(x) = \mathbb{E}\left[Y_i(0) \mid X_i = x\right].$$

Can we estimate the average treatment effect $\mathbb{E}\left[Y_i(1) - Y_i(0)\right]$? Only if we assume something like $\mathbb{E}\left[Y_i(1) - Y_i(0) \mid X_i = x\right]$ is constant. The causal estimand is the ATE at the cutoff:

$$\theta = \mathbb{E}\left[Y_i(1) - Y_i(0) \mid X_i = c\right].$$

We will use nonparametrics to estimate the conditional response functions $\mu_{(1)}(x) \coloneqq \mathbb{E}\left[Y_i(1) \mid X_i = x\right]$ and $\mu_{(0)}(x) \coloneqq \mathbb{E}\left[Y_i(0) \mid X_i = x\right]$ near the cutoff c. We can assume that both $\mu_{(1)}, \mu_{(2)}$

are continuous at c, and as a result, write

$$\begin{aligned}\theta &= \mathbb{E}\left[Y_i(1) - Y_i(0) \mid X_i = c\right] \\ &= \mu_{(1)}(c) - \mu_{(0)}(c) \\ &= \lim_{x \downarrow c} \mu_{(1)}(x) - \lim_{x \uparrow c} \mu_{(0)}(x)\end{aligned}$$

with the estimators $\hat{\mu}_{(1)}, \hat{\mu}_{(0)}$ from nonparametric regression.

This continuity assumption of μ at c can be upgraded to something like Hölder continuity.

Estimating the treatment effect at the cutoff with RDD is a bit problematic because we have extra boundary bias. In practice, we can do local linear regression instead.

What happens if X_i is discrete? Recall the test scores example: the highest score of any individual in the control group is 9, which we may consider to be too far from the cutoff score of 10. There are several ways to deal with this:

- We can assume X_i has continuous density at c
- Bias-aware approach: use a continuity assumption to get an upper bound on the possible bias, and widen confidence intervals accordingly. When we assume $\mu_{(1)}(\cdot) \in \Sigma(1, c)$ but X_i is discrete, we are assuming there exists $\tilde{\mu}_{(1)}$ that is an extension of $\mu_{(1)}(c)$ and in $\Sigma(1, c)$.

5.2 Estimating Treatment Effects in a RCT

Consider an RCT in which $W_i \perp\!\!\!\perp (X_i, Y_i(0), Y_i(1))$ and $p = \mathbb{P}\left[W_i = 1\right] \in (0, 1)$.

Suppose a medication affects one cohort in the population differently than another. The ATE is misleading because it does not capture how treatment affects the different cohorts differently.

We would rather estimate the conditional average treatment effect:

$$\tau(x) = CATE(x) \coloneqq \mathbb{E}\left[Y_i(1) - Y_i(0) \mid X_i = x\right] = \mu_{(1)}(x) - \mu_{(0)}(x).$$

By the LIE, we can recover the ATE from the CATEs:

$$ATE = \mathbb{E}\left[CATE(X_i)\right].$$

Definition 5.2.1 (t-learner).

For some generic nonparametric method $\mathscr{M}$, we can estimate $\tau(x)$ by estimating

$$\begin{aligned}\hat{m}_{(1)}(\cdot) &= \mathscr{M}\left[Y_i \sim X_i, i \in \mathscr{D}_1 = \{i : W_i = 1\}\right], \\ \hat{m}_{(0)}(\cdot) &= \mathscr{M}\left[Y_i \sim X_i : i \in \mathscr{D}_0 = \{i : W_i = 0\}\right]'\end{aligned}$$

then putting

$$\tau(x) = \hat{m}_{(1)}(x) - \hat{m}_{(2)}(x).$$

$\mathscr{M}$ is learning on $\mathscr{D}_1$ the quantity $\mathbb{E}\left[Y_i \mid X_i, W_i = 1\right]$, which by SUTVA then independence on W_i and $Y_i(1)$, is equal to $\mathbb{E}\left[Y_i(1) \mid X_i = x\right]$.

The regression method $\mathscr{M}$ may differ between the treatment and control groups, for example if their sample sizes differ sharply.

Definition 5.2.2 (s-learner).

An alternative is to estimate

$$\hat{m}(\cdot) = \mathscr{M}\left[Y_i \sim (X_i, W_i) \text{ for } i = 1, \ldots, n\right].$$

Then we can estimate

$$\hat{\tau}(x) = \hat{m}(x, 1) - \hat{m}(x, 0).$$

Remark 5.2.3. Both t- and s-learning can perform poorly.

If we have few treated[1] units and $\tau(\cdot)$ is not complicated (in the sense that it is smoother than $\mu_{(0)}(x)$), then the t-learner would be bad.

Definition 5.2.4 (Simplified x-learner).

Still consider

$$\hat{\mu}_{(0)}(\cdot) = \mathscr{M}\left[Y_i \sim X_i : i \in \mathscr{D}_0 = \{i : W_i = 0\}\right].$$

Then write

$$\tilde{Y}_i = Y_i - \hat{\mu}_{(0)}(X_i)$$

for $i \in \mathscr{D}_1$, and put the estimator

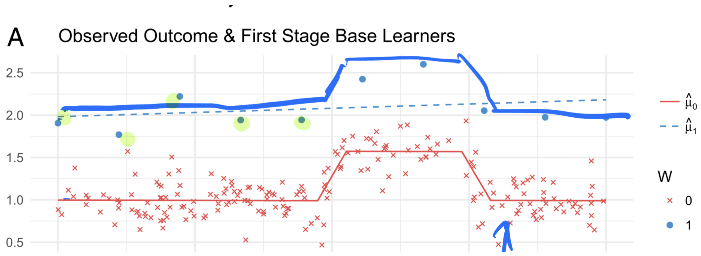
$$\hat{\tau}(\cdot) = \mathscr{M}\left[\tilde{Y}_i \sim X_i \text{ for } i \in \mathscr{D}_1\right]$$

for treated units. We see that

$$\begin{aligned}
\mathbb{E}\left[\tilde{Y}_i \mid X_i = x, W_i = 1\right] &= \mathbb{E}\left[Y_i - \hat{\mu}_{(0)}(X_i) \mid X_i = x, W_i - 1\right] \\
&= \mathbb{E}\left[Y_i \mid X_i = x, W_i = 1\right] - \hat{\mu}_{(0)}(x) \\
&= \mu_{(1)}(x) - \hat{\mu}_{(0)}(x) \\
&\approx \mu_{(1)}(x) - \mu_{(0)}(x) = \tau(x).
\end{aligned}$$

Now we are directly estimating the object we care about. We can impose smoothness assumptions on $\tau(\cdot)$.

Figure 5.1: Comparison of the t and x-learners.



Here, we see an example of sparse treatment, where the x-learner would fare

[1] In class, he said control but I think "treated" is more consistent with the example in the slides.

much better than the t-learner.

Definition 5.2.5 (Full x-learner).

Call the above estimator $\hat{\tau}_1$ and do the same thing with roles of $W_i = 1$ and $W_i = 0$ reversed to get $\hat{\tau}_0$. Then put

$$\hat{\tau}(x) = (1-p)\hat{\tau}_1(x) + p\hat{\tau}_0(x).$$

More precisely, put

$$\begin{aligned}
\hat{m}_0(\cdot) &= \mathscr{M}_1\left[Y_i \sim X_i, i \in \mathscr{D}_0\right] \\
\hat{\tau}_0(\cdot) &= \mathscr{M}_2\left[\underbrace{Y_i - \hat{m}_0(X_i)}_{\tilde{Y}_i} \sim X_i, i \in \mathscr{D}_1\right] \\
\hat{m}_1 &= \mathscr{M}_3\left[Y_i \sim X_i, i \in \mathscr{D}_1\right] \\
\hat{\tau}_1(\cdot) &= \mathscr{M}_4\left[\hat{m}_1(X_i) - Y_i \sim X_i, i \in \mathscr{D}_0\right] \\
\hat{\tau}(x) &= (1-p)\hat{\tau}_0(x) + p\hat{\tau}_1(x).
\end{aligned}$$

The simplified x-learner is useful with few treatment observations and many controls. The full x-learner is useful in the opposite regime.

$\tilde{Y}_i$ imputes the missing potential outcome, so its conditional expectation is approximately $\tau(X_i)$. The same logic applies to $\hat{\tau}_1$.

Definition 5.2.6 (F-learner).

Put

$$\begin{aligned}
\tilde{Y}_i &= \frac{W_i Y_i}{p} - \frac{(1-W_i)\,Y_i}{1-p} \\
&= \begin{cases} \frac{Y_i}{p}, & W_i = 1 \\ -\frac{Y_i}{1-p}, & W_i = 0 \end{cases}.
\end{aligned}$$

Then use the estimator

$$\hat{\tau}(\cdot) = \mathscr{M}\left[\tilde{Y}_i \sim X_i \text{ for all } i = 1, \ldots, n\right].$$

We can check

$$\begin{aligned}
\hat{\tau}(x) &= \mathbb{E}\left[\tilde{Y}_i \mid X_i\right] \\
&= \mathbb{E}\left[\frac{W_i Y_i}{p} - \frac{(1-W_i)Y_i}{1-p} \mid X_i\right] \\
&= \mathbb{E}\left[\frac{W_i Y_i}{p} - \frac{(1-W_i)Y_i}{1-p} \mid X_i, W_i = 1\right] \mathbb{P}\left[W_i = 1\right] \\
&\quad + \mathbb{E}\left[\frac{W_i Y_i}{p} - \frac{(1-W_i)Y_i}{1-p} \mid X_i, W_i = 0\right] \mathbb{P}\left[W_i = 0\right] \\
&= \mathbb{E}\left[\frac{Y_i}{p} \mid X_i, W_i = 1\right] p - \mathbb{E}\left[\frac{Y_i}{1-p} \mid X_i, W_i = 0\right] (1-p) \\
&= \mathbb{E}\left[\frac{Y_i(1)}{p} \mid X_i\right] p - \mathbb{E}\left[\frac{Y_i(0)}{1-p} \mid X_i\right] (1-p) \\
&= \mathbb{E}\left[Y_i(1) - Y_i(0) \mid X_i\right] = \tau(x).
\end{aligned}$$

Remark 5.2.7. The F-learner is exactly conditionally unbiased. However, if p is close to 0 or 1, $\operatorname{Var}\left(\tilde{Y}_i \mid X_i\right)$ could be very large.

5.3 Meta-Learners in Observational Studies

Let's move from RCTs to observational studies, so we do not randomly assignment treatment, nor do we know $p = \mathbb{P}\left[W_i = 1\right]$.

Notation 5.3.1.

We will now notate

$$e \coloneqq p = \mathbb{P}\left[W_i = 1\right] \in (0, 1).$$

Assumption 5.3.2.

We wil have standing assumptions of unconfoundedness and overlap.

Definition 5.3.3 (Unconfoundedness).

The treatment is unconfounded if it is independent of potential outcomes, conditional on covariates:

$$W_i \perp\!\!\!\perp (Y_i(0), Y_i(1)) \mid X_i.$$

Definition 5.3.4 (Propensity score).

Define the propensity score

$$e(x) \coloneqq \mathbb{P}\left[W_i = 1 \mid X_i = x\right].$$

In a non-RCT setting, $e(x)$ is generally unknown.
We say that there is overlap in the study if $e(x) \in (0,1)$ for every x.

The t-learner makes sense still, and the estimates $\hat{\tau}_0$, $\hat{\tau}_1$ in the x-learner also make sense. Ideally, we would write

$$\tau(x) = (1 - e(x))\hat{\tau}_0(x) + e\hat{\tau}_1(x),$$

RDD typically does not satisfy overlap, since treatment is determined by whether the running variable crosses the cutoff.

but we don't know $e(x) = \mathbb{P}\left[W_i = 1 \mid X_i = x\right]$. We can estimate e by regressing W_i on the covariates:

$$\hat{e}(\cdot) = \mathscr{M}\left[W_i \sim X_i, \forall\ i\right],$$

then estimate

$$\hat{\tau}(x) = (1 - \hat{e}(x))\hat{\tau}_0(x) + \hat{e}(x)\hat{\tau}_1(x).$$

With the F-learner, we can put

$$\tilde{Y}_i = \frac{y_i W_i}{\hat{e}(X_i)} - \frac{Y_i(1 - W_i)}{1 - \hat{e}(X_i)},$$

then estimate

$$\hat{\tau}(\cdot) = \mathscr{M}\left[\tilde{Y}_i \sim X_i, \forall\ i\right].$$

However, we lose the main appeal of the F-learner, in that $\mathbb{E}\left[\tilde{Y}_i \mid X_i\right] \neq \tau(X_i)$. Furthermore, if $e(X_i) \approx 0$ for some X_i, then the F-learner may blow up.[2]

We lose the unbiasedness of the F-learner because the propensity score $e(x)$ is no longer known (as it was in the RCT) and we estimate it with $\hat{e}$.

These learners are actually not so great for use in observational studies. We introduce the Doubly Robust learner.

Remark 5.3.5. For intuition, think of $\tilde{m}_0(\cdot), \tilde{m}_1(\cdot), \tilde{e}(\cdot)$ as fixed functions. Then analogously to as in the t-learner (with the new addition of debiasing terms), consider

$$\tilde{Y}_i(\tilde{m}_1, \tilde{m}_0, \tilde{e}) \coloneqq \tilde{m}_1\left(X_i\right) - \tilde{m}_0(X_i) + \frac{W_i\left(Y_i - \tilde{m}_1(X_i)\right)}{\tilde{e}(X_i)} - \frac{(1 - W_i)(Y_i - \tilde{m}_0(X_i))}{1 - \tilde{e}(X_i)}.$$

The first two terms resemble the t-learner, while the second two terms resemble the F-learner, but centered around the regression functions.

Suppose $\tilde{m}_1 = m_1, \tilde{m}_0 = m_0$, but $\tilde{e}$ is arbitrary. Then

$$A_i \coloneqq m_1(X_i) + \frac{W_i(Y_i - m_1(X_i))}{\tilde{e}(X_i)}, \qquad B_i \coloneqq m_0(X_i) + \frac{(1 - W_i)(Y_i - m_0(X_i))}{1 - \tilde{e}(X_i)}.$$

A short conditional-expectation calculation gives

$$\mathbb{E}\left[A_i \mid X_i\right] = m_1(X_i), \qquad \mathbb{E}\left[B_i \mid X_i\right] = m_0(X_i).$$

Thus $\mathbb{E}\left[\tilde{Y}_i(m_1, m_0, \tilde{e})\right] = \text{CATE}$.

[2] In practice, we may use a heuristic to clip $\hat{e} \in [\epsilon, 1 - \epsilon]$.

Now suppose $\tilde{m}_1, \tilde{m}_0$ are arbitrary but the propensity is true in $\tilde{e} = e$. Then

$$\tilde{A}_i := \tilde{m}_1(X_i) + \frac{W_i(Y_i - \tilde{m}_1(X_i))}{e(X_i)}, \qquad \tilde{B}_i := \tilde{m}_0(X_i) + \frac{(1 - W_i)(Y_i - \tilde{m}_0(X_i))}{1 - e(X_i)}.$$

Again, conditioning on X_i gives

$$\mathbb{E}\left[\tilde{A}_i \mid X_i\right] = m_1(X_i), \qquad \mathbb{E}\left[\tilde{B}_i \mid X_i\right] = m_0(X_i).$$

Thus $\mathbb{E}\left[\tilde{Y}_i(\tilde{m}_1, \tilde{m}_0, e)\right] = \text{CATE}$.

If we know perfectly the conditional means of the potential outcomes (m_1, m_0) or the propensity score (e), then the expectation of $\tilde{Y}_i$ would give the CATE. This motivates the DR-Learner.

Definition 5.3.6 (DR-Learner).

Put

$$\begin{aligned}
\hat{m}_0(\cdot) &= \mathscr{M}_1\left[Y_i \sim X_i, i : W_i = 0\right] \\
\hat{m}_1(\cdot) &= \mathscr{M}_2\left[Y_i \sim X_i, i : W_i = 1\right] \\
\hat{e}(\cdot) &= \mathscr{M}_2\left[W_i \sim X_i, \forall\ i\right] \\
\hat{\tau}(\cdot) &= \mathscr{M}_4\left[\tilde{Y}_i\left(\hat{m}_1, \hat{m}_0, \hat{e}\right) \sim X_i, \forall\ i\right].
\end{aligned}$$

Aside 5.3.7. This is better than our other learners for observations studies. The first-order errors cancel, so the remaining bias is second-order: conditioning on $X_i = x$, it is roughly on the scale of

$$(\hat{e}(x) - e(x))\left((\hat{m}_1(x) - m_1(x)) + (\hat{m}_0(x) - m_0(x))\right).$$

Remark 5.3.8. There is one remaining problem: overfitting, which we can mitigate by cross-fitting.

Split the data into two folds I_1, I_2 as in cross-validation (each contains treatment and controls). Then we estimate separately on the two folds $\hat{m}_0^{I_1}, \hat{m}_1^{I_1}, \hat{e}^{I_1}$ and $\hat{m}_0^{I_2}, \hat{m}_1^{I_2}, \hat{e}^{I_2}$. Then for each $i \in I_1$, we put

$$\tilde{Y}_i = \tilde{Y}_i\left(\hat{m}_1^{I_2}, \hat{m}_0^{I_2}, \hat{e}^{I_2}\right),$$

and for $i \in I_2$,

$$\tilde{Y}_i = \tilde{Y}_i\left(\hat{m}_0^{I_1}, \hat{m}_1^{I_1}, e^{I_1}\right).$$

Then we pool and estimate

$$\hat{\tau}(\cdot) = \mathscr{M}_5\left[\tilde{Y}_1 \sim X_i, \forall\ i\right].$$

The main takeaway is that we can solve lots of prediction problems and combine the results in an intelligent way to solve problems outside prediction.

Miscellany 6

6.1 Global Linear Regression and Splines

Consider

$$\min_{m \in \mathscr{M}_n} \frac{1}{n} \sum_{i=1}^{n} \left(Y_i - m(X_i)\right)^2 .$$

This is the usual least squares objective, but we restrict m to lie in some model class $\mathscr{M}_n$. This is often called a *series estimator*: we choose a finite-dimensional family of functions and then fit that family by least squares. The simplest example would be

Typically $d_n \xrightarrow{n\to\infty} \infty$ slowly.

$$\mathscr{M}_n \coloneqq \{\text{Polynomials of degree at most } d_n\} ,$$

so we are considering

$$\min_{\{\alpha_j\}} \frac{1}{n} \sum_{i=1}^{n} \left(Y_i - \sum_{j=0}^{d_n} \alpha_j X_i^j \right)^2 .$$

Equivalently, $m(x)$ is parameterized as a polynomial in x, and we fit the coefficients by least squares. For fixed d_n, this is just an ordinary linear regression with regressors $1, X_i, \ldots, X_i^{d_n}$.

By Stone–Weierstrass, we can approximate a $C\left([0,1]\right)$ function by a polynomial P within any ϵ error:

$$\sup_{x \in [0,1]} |m(x) - P(x)| < \epsilon.$$

So if d_n is allowed to grow with n, the approximation bias can shrink. The tradeoff is that larger d_n means more coefficients to estimate, hence more variance. The usual heuristic is:

- $d_n \to \infty$ so the model class becomes rich enough to approximate the truth.
- $d_n/n \to 0$ so we are not estimating too many coefficients relative to the sample size.

Under this balance, we expect consistency in

$$\frac{1}{n} \sum_{i=1}^{n} \left(m(X_i) - \hat{p}_n(X_i)\right)^2 \xrightarrow[n\to\infty]{} 0.$$

Here the conditions $d_n \to \infty$ and $d_n/n \to 0$ say that the approximation error goes to 0 while the estimation error does not blow up.

This estimator is a linear smoother. The reason is simpler than the local-polynomial analogy: for any fixed basis $h(x)$, least squares gives

$$\hat{m}(x_0) = h(x_0)^T \left(X^T X\right)^{-1} X^T Y = \sum_{i=1}^{n} \gamma_i(x_0) Y_i,$$

so the fitted value at x_0 is linear in the responses.

This estimator is a bad idea if we're doing regression discontinuity design (Imbens and Gelman, 2019).[1] The issue is that a global polynomial uses observations far from the cutoff to determine the fit near the cutoff, so the local causal estimand can be dominated by global shape assumptions.

What is a better choice for $\mathscr{M}_n$? A natural choice would be a class of splines. Roughly, a *spline* is a function formed by gluing together low-degree polynomials on adjacent intervals, usually with continuity conditions imposed at the join points. Those join points are called *knots*. For us, the main examples are piecewise linear and piecewise cubic splines. Splines are still global least-squares estimators, but they allow the slope or curvature to change at a prescribed set of knots. This gives more local flexibility than a single polynomial.

See Elements of Statistical Learning, Chapter 5.

Oftentimes, $\mathscr{M}_n$ is a vector space, so it suffices to define a basis. For example, for a piecewise constant function with discontinuities at ξ_1, ξ_2, we could define the basis

$$h(x) = \begin{pmatrix} \mathbb{1}_{x<\xi_1} \\ \mathbb{1}_{x\in[\xi_1,\xi_2)} \\ \mathbb{1}_{x\geq\xi_2} \end{pmatrix}.$$

If we wanted to take $\mathscr{M}_n$ as the class of piecewise linear functions with the same discontinuity points, we could take

$$h(x) = \begin{pmatrix} \mathbb{1}_{x<\xi_1} \\ \mathbb{1}_{x\in[\xi_1,\xi_2)} \\ \mathbb{1}_{x\geq\xi_2} \\ x\mathbb{1}_{x<\xi_1} \\ x\mathbb{1}_{x\in[\xi_1,\xi_2)} \\ x\mathbb{1}_{x\geq\xi_2} \end{pmatrix}.$$

For a continuous piecewise linear function with two knots (kinks), we have that the basis is of dimension 4, as we can write

$$h(x) = \begin{pmatrix} 1 \\ x \\ (x-\xi_1)_+ \\ (x-\xi_2)_+ \end{pmatrix}.$$

The first argument controls a global intercept, the second a global slope, and the third and fourth control changes in the slope in the two remaining segments.

The point of this truncated-power basis is that each new basis function only changes the fit after its knot, so knots localize the flexibility.

Recall $(t)_+ = \max\{t, 0\}$, so $(x-\xi)_+$ only affects the fit to the right of the knot ξ.

Piecewise cubic polynomials are most used in practice. If we allow for discontinuity, we have 12 parameters. Continuity reduces to 10, C^1 has 8, and C^2 forces just 6 parameters. C^2 piecewise cubic polynomials are called cubic splines and are the

[1] Perhaps don't predoc with Greenstone.

most common choice. Cubics are flexible enough to bend, but the C^2 constraint prevents the fit from changing too abruptly from one segment to the next.

A *B-spline basis* is another basis for the same spline space. Unlike the truncated-power basis above, each B-spline basis function is nonzero only on a small interval around a few adjacent knots, so B-splines are more local and numerically more stable. The important point is that B-splines do not define a different class of functions; they are just a different basis for writing splines.

An example of a B-spline basis is a basis consisting of Gaussian densities with different means.

Lastly, natural splines enforce linearity before the first knot and after the last knot. This removes the most unstable tail behavior of ordinary cubic splines. If we count the two boundary knots together with the interior knots, then a natural spline with n total knots has dimension n; equivalently, with K interior knots, the dimension is $K + 2$.

For computational use in R, `bs()` constructs a B-spline basis and `ns()` constructs a natural spline basis.

Consider the regression spline $\texttt{lm}(y \sim ns(x, \ldots))$. Is this a linear smoother? With the problem

A regression spline is not necessarily limited to a natural spline.

$$\min_{\alpha_1,\ldots,\alpha_d} \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - \sum_{j=1}^{d}\alpha_j h_j(X_i)\right)^2,$$

Regression splines work well and are very easy to implement once we determine the basis (which can be done with software)

the key point is that we are minimizing a quadratic objective over coefficients, so the fitted values will be linear in Y. We write the design matrix

$$X = \begin{pmatrix} h_1(X_1) & \cdots & h_d(X_1) \\ \vdots & \ddots & \vdots \\ h_1(X_n) & \cdots & h_d(X_n) \end{pmatrix},$$

and put

$$\hat{\alpha} = \left(X^T X\right)^{-1} X^T Y$$

to get

$$\hat{m}(x) = \sum_{j=1}^{d}\hat{\alpha}_j h_j(x) = \sum_{i=1}^{n}\gamma_i(x) Y_i.$$

More explicitly, $\hat{m}(x) = h(x)^T(X^TX)^{-1}X^TY$, so the weights γ_i depend on x and the X_i, but not on Y_i.

The main challenge with regression splines is that the knot locations are somewhat arbitrary. By default, R typically fixes the dimension and then places the knots at quantiles of the observed X_i. This is a reasonable generic choice, but ideally we would want more knots where the regression function changes quickly and fewer where it is nearly linear.

6.2 Penalized Least Squares

Consider penalized least squares, where we solve

$$\min_{m} \frac{1}{n}\sum_{i=1}^{n}\left(Y_i - m(X_i)\right)^2 + \lambda\,\mathrm{Pen}(m), \tag{6.1}$$

where the minimum is taken over all functions m. The philosophy is: instead of choosing a small model class by hand, we allow a very large class of functions and penalize rough ones.

A smoothing spline uses the penalty $\lambda \int_0^1 (m''(x))^2\, dx$, where the penalty is $+\infty$ if the second derivative does not exist on a nonnull set. This penalty charges curvature: the more the function bends, the larger the penalty. As $\lambda \to \infty$, we heavily penalize curvature, so $\hat{m}$ is forced toward a linear function and minimizes $\frac{1}{n}\sum_{i=1}^{n}(Y_i - \alpha - \beta X_i)^2$, despite the minimization being over all functions. As $\lambda \to 0$, we fit the data more aggressively.

Theorem 6.2.1 (Representer).

The minimization (6.1) has a solution that is a natural spline with knots at the observed data points X_i. In other words, we do not have to choose the knots in advance: the representer theorem says the optimizer can be written using the sample points themselves as knots. Computation is on the order of $O(n)$, which is very quick with R functions like `smooth.spline`.

This theorem is the key computational fact: even though we wrote down an infinite-dimensional optimization problem, the optimizer lives in a finite-dimensional spline space. By Silverman (1984), there is an exact expression for the equivalent kernel of the smoothing spline, so smoothing splines can also be viewed through the linear-smoother lens.

Considering the class

$$\mathscr{S}(2, C) = \left\{ m : m \in C^2,\ \int (m''(x))^2\, dx \le C^2 \right\},$$

the smoothing spline achieves the minimax risk for integrated MSE for $X \sim \mathbb{P}_X$:

$$\mathbb{E}\left[(\hat{m}(X) - m(X))^2 \right] = n^{-\frac{4}{5}}.$$

This is the same one-dimensional nonparametric rate for two derivatives ($s = 2$) we have seen before: $n^{-2s/(2s+1)} = n^{-4/5}$.

Example 6.2.2.

If there are no interior knots, then a natural cubic spline is just a line, so a basis is $h(x) = (1,\ x)'$. With one interior knot ξ, we add one truncated-cubic term to capture the extra bend after ξ. In practice, software builds this basis for us, so the main conceptual point is that "more knots" means "more local flexibility."

6.3 Not Linear Smoothers

Everything we've seen so far is a linear smoother. Is there any loss in restricting study to linear smoothers?

Let LR be the minimax optimal rate among linear smoothers, and let MR be the minimax optimal rate among all possible estimators. The question is whether allowing nonlinear estimators can improve the rate.

Considering the class $\sup_{x\in[0,1]} |m''(x)| \le C$, the minimax rate is

$$\mathrm{MR} = n^{-\frac{4}{5}} = \mathrm{LR}.$$

With the broader class $\mathscr{S}(2, C)$, we still have

$$\mathrm{MR} = n^{-\frac{4}{5}} = \mathrm{LR},$$

where the smoothing spline achieves the optimal rate.

However, with the even broader class $\mathrm{TV}(2, C) = \{m : \int |m''(x)|\, dx \le C\}$, Donoho–Johnstone (1998) showed $\mathrm{MR} = n^{-\frac{4}{5}} < n^{-\frac{3}{4}} = \mathrm{LR}$. The total variation class allows functions whose curvature is concentrated in a few places, and linear smoothers cannot adapt sharply enough to that kind of spatial inhomogeneity.

Which estimator achieves the $n^{-\frac{4}{5}}$ rate? While the smoothing spline resulted from solving

$$\min_m \left\{ \frac{1}{n} \sum_{i=1}^n (Y_i - m(X_i))^2 + \lambda \int_0^1 (m''(x))^2\, dx \right\},$$

we can put the locally adaptive spline as

$$\min_m \left\{ \frac{1}{n} \sum_{i=1}^n (Y_i - m(X_i))^2 + \lambda \int_0^1 |m''(x)|\, dx \right\},$$

which is not a linear smoother.[2] Replacing the squared penalty by an absolute-value penalty makes the estimator adaptive: it can spend its curvature budget in a few difficult regions instead of spreading it evenly everywhere. There is no representer theorem nor linear computational time, so the locally adaptive spline is harder to compute and not as widely used.

Heuristic: the $\int |m''|$ penalty encourages m'' to be sparse, i.e. a small number of "kinks" in the second derivative.

If we sort the covariates so that $X_1 \le \cdots \le X_n$, a computable approximation replaces the integral penalty by a discrete second-difference penalty. By Boyd et al. (2010) and more importantly Ryan, Tibshirani (2014), people have considered the approximation to the locally adaptive spline of

$$\min_m \left\{ \frac{1}{n} \sum_{i=1}^n (Y_i - m(X_i))^2 + \lambda \sum_{i=1}^{n-2} |m(X_i) - 2m(X_{i+1}) + m(X_{i+2})| \right\},$$

which just depends on $m(X_1), \ldots, m(X_n)$ and is a finite-dimensional optimization problem. Under some conditions, this is asymptotically equivalent to locally adaptive splines.

6.4 Classification and Regression Trees (CART)

In step 1 of the CART algorithm, we find a cutoff c and fit a piecewise constant function on

$$I_L = \{x : x \le c\}, \quad I_R = \{x : x > c\},$$

putting $\hat{m}(x)$ equal to the average of the Y_i in the region containing x. The cutoff c is chosen to minimize the residual sum of squares after the split. In step 2, we iterate on this and further divide the subregions. So CART is a greedy recursive partitioning method: keep splitting the feature space into rectangles (or intervals in one dimension), then average within the terminal leaves.

[2] This fact is given to us. The locally adaptive spline is like the lasso regression and the smoothing spline like the ridge regression.

CART is in fact not a linear smoother. While we can write

$$\hat{m}(x) = \sum_{i=1}^{n} \gamma_i(x) Y_i, \quad \gamma_i(x) = \begin{cases} \frac{1}{\#\{j : X_j \in I(x)\}} & X_i \in I(x) \\ 0 & X_i \notin I(x) \end{cases}$$

for $I(x)$ the terminal leaf that x falls into, $I(x)$ really depends on all the Y_is, which is not allowed for a linear smoother.

Honest trees are variants that are linear smoothers.

Definition 6.4.1 (Honest trees, Athey and Imbens (2016)).

Suppose we have observations $(Y_i, X_i)_{1:n}$. We split the observations into two folds F_1, F_2, learn on the first fold F_1, and use F_2 to make predictions for the leaves.

Conditioning on F_1, the tree structure is fixed, so $\hat{m}$ is a linear smoother conditional on F_1. The sample split is what separates "choosing the weights" from "averaging the responses with those weights."

Remark 6.4.2. Some disadvantages of CART include:

- Different leaves can have very different sample sizes, so variance can vary a lot across leaves.
- It produces a very unstable estimator: changing one data point can lead to a very different tree.
- Because it is piecewise constant, it can miss smooth trends and often predicts worse than smoother methods.
- Since CART is analogous to a regressogram, there is boundary bias.
- The estimator is not smooth.

On the other hand, without too many splits, the tree is quite interpretable.

Definition 6.4.3 (Random forests, Breiman (2001)).

Random forests fit a lot of trees $T^1, \ldots, T^B$ on subsamples of all observations. We introduce additional randomness:

- For each tree, we first resample with replacement the subsample observations. (A single data point can appear multiple times.)
- For each split, randomly pick only a subset of features and consider only those as candidates to split.

The final $\hat{m}(\cdot)$ is the average of all tree predictions.

Random forests address the shortcomings of CART by averaging many unstable trees, which substantially reduces variance and usually improves prediction accuracy. While there are honest random forests that use many honest trees, the usual random forest is still not a linear smoother because the leaves themselves depend on the observed responses.

Let's proceed as if a random forest were a linear smoother to gain intuition: for

$$\hat{m}(x) = \frac{1}{B}\sum_{t=1}^{B}\hat{m}_t(x) = \frac{1}{B}\sum_{t=1}^{B}\frac{1}{|I_t(x)|}\sum_{i=1}^{n}\mathbb{1}_{i\in I_t(x)}Y_i = \sum_{i=1}^{n}\left(\underbrace{\frac{1}{B}\sum_{t=1}^{B}\frac{\mathbb{1}_{i\in I_t(x)}}{|I_t(x)|}}_{\gamma_i(x)}\right)Y_i,$$

where $I_t(x)$ is the leaf of tree t that contains x. This suggests viewing a forest as a *data-adaptive kernel*: points that frequently land in the same leaves as x receive large weight.

What if we tried to do random forests using locally linear regression, using a kernel coming from the random forest? That is the basic motivation behind local linear forests.

6.5 More on Meta-Learners: Using Regression in Other Applications

Regression ideas show up in other problems too. Suppose $C_i \sim \text{Bernoulli}(\frac{1}{2})$, and the feature distribution depends on the class:

$$X_i \mid C_i = 1 \sim r(\cdot) \text{ (Density 1)}$$
$$X_i \mid C_i = 0 \sim q(\cdot) \text{ (Density 2).}$$

Our goal is to estimate the density ratio $r(x)/q(x)$.

We have some options:

(i) (T-learner, ish) We can estimate $\hat{r}(x)$ based on $\{X_i : C_i = 1\}$ and $\hat{q}(x)$ based on $\{X_i : C_i = 0\}$, then put our estimate as $\hat{r}(x)/\hat{q}(x)$.

What if their ratio is simpler than the two densities separately? Then it can be better to reduce the problem to a regression or classification problem.

(ii)

$$\begin{aligned}
\mathbb{P}[C_i = 1 \mid X_i] &\overset{\text{Bayes}}{=} \frac{\mathbb{P}[X_i \mid C_i = 1]\,\mathbb{P}[C_i = 1]}{\mathbb{P}[X_i]} \\
&= \frac{\frac{1}{2}r(X_i)}{\frac{1}{2}\left(r(X_i) + q(X_i)\right)} \\
&= \frac{r(X_i)}{r(X_i) + q(X_i)} \\
1 - \mathbb{P}[C_i = 1 \mid X_i] &= \frac{q(X_i)}{r(X_i) + q(X_i)} \\
\frac{\mathbb{P}[C_i = 1 \mid X_i]}{1 - \mathbb{P}[C_i = 1 \mid X_i]} &= \frac{r(X_i)}{q(X_i)}.
\end{aligned}$$

Since $\mathbb{P}[C_i = 1 \mid X_i] = \mathbb{E}[C_i \mid X_i]$, if we have a way of estimating $\hat{m}$ from

$C_i \sim X_i$, then we can put

$$\widehat{\left(\frac{r(x)}{q(x)}\right)} = \frac{\hat{m}(x)}{1 - \hat{m}(x)}.$$

So density-ratio estimation can be converted into conditional-mean estimation, which may be easier to attack with the regression tools from the course.

Now we pivot to consider $X_i \sim f$ and want to estimate the score

$$s(x) = \frac{\partial}{\partial x} \ln\left(f(x)\right) = \frac{f'(x)}{f(x)}.$$

Again, we could separately estimate f' and f, but that is often unstable. A better and more modern approach is to add Gaussian noise: let $\epsilon_i \sim N(0, \tau^2)$ and define $Z_i = X_i + \epsilon_i$. If K is the Gaussian kernel, then the density of Z_i is

$$f_Z(z) = \mathbb{E}\left[\frac{1}{\tau} K\left(\frac{X_i - z}{\tau}\right)\right] \overset{\tau \text{ small}}{\approx} f(z),$$

for K the Gaussian kernel.

Alternatively, this is the expected KDE with Gaussian kernel bandwidth τ, or equivalently the density of the convolution of ϵ_i and X_i.

Heuristically, if τ is small, then Z is just a slightly blurred version of X, so the score of f_Z should be close to the score of f.

Proposition 6.5.1 (Tweedie's formula, Eddington (1926)).

We have that

$$\mathbb{E}\left[X_i \mid Z_i = z\right] = z + \tau^2 \frac{\partial}{\partial z} \ln\left(f_Z(z)\right).$$

With this, we can estimate the score of the noisy density by regressing X_i on Z_i to get $\hat{m}$, and then put

$$\hat{s}_Z(z) = \frac{\hat{m}(z) - z}{\tau^2}.$$

When τ is small, this provides an approximation to the score of the original density f.

This is called *denoising score matching*, which underlies score-based diffusion models.